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(54) **ON-LINE GAMING MONITORING SYSTEMS AND USER INTERFACES**

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(57) **ABSTRACT**

Systems and methods described herein include a video gaming server configured to communicate with a video gaming device of a first user. A risk module is coupled to the video gaming server. The risk module is configured to: monitor online activities of the first user when the first user operates the video gaming device; determine that the first user has encountered an online risk while operating the video gaming device; and cause a notification of the online risk to be sent to a second device operated by a second user, the first and second users being associated with each other by a common account. The notification presents the second user with one or more options for responding to the online risk encountered by the first user.

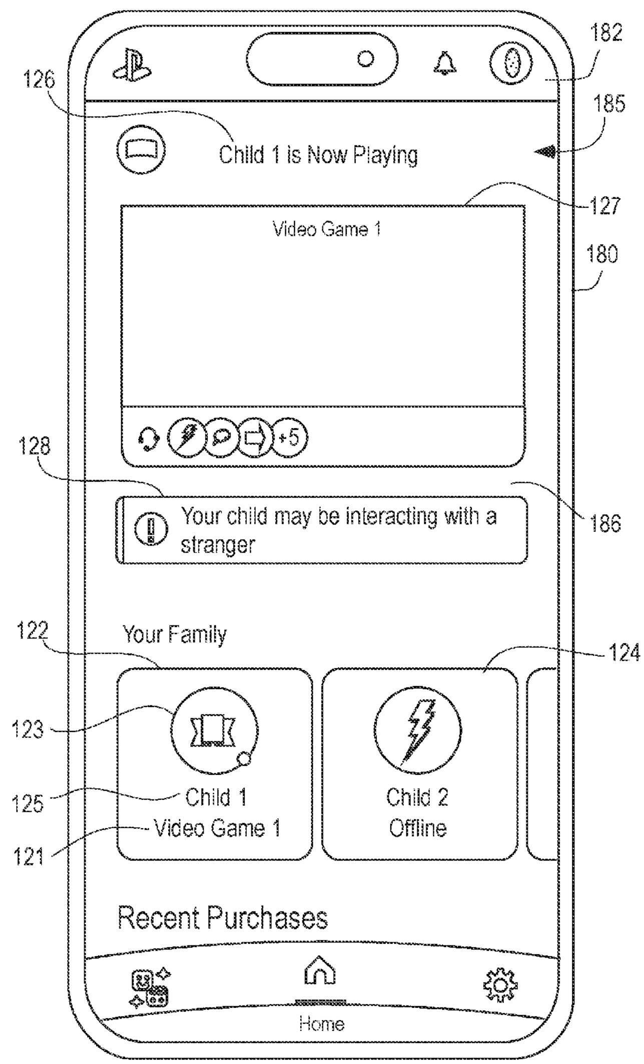


FIG. 1

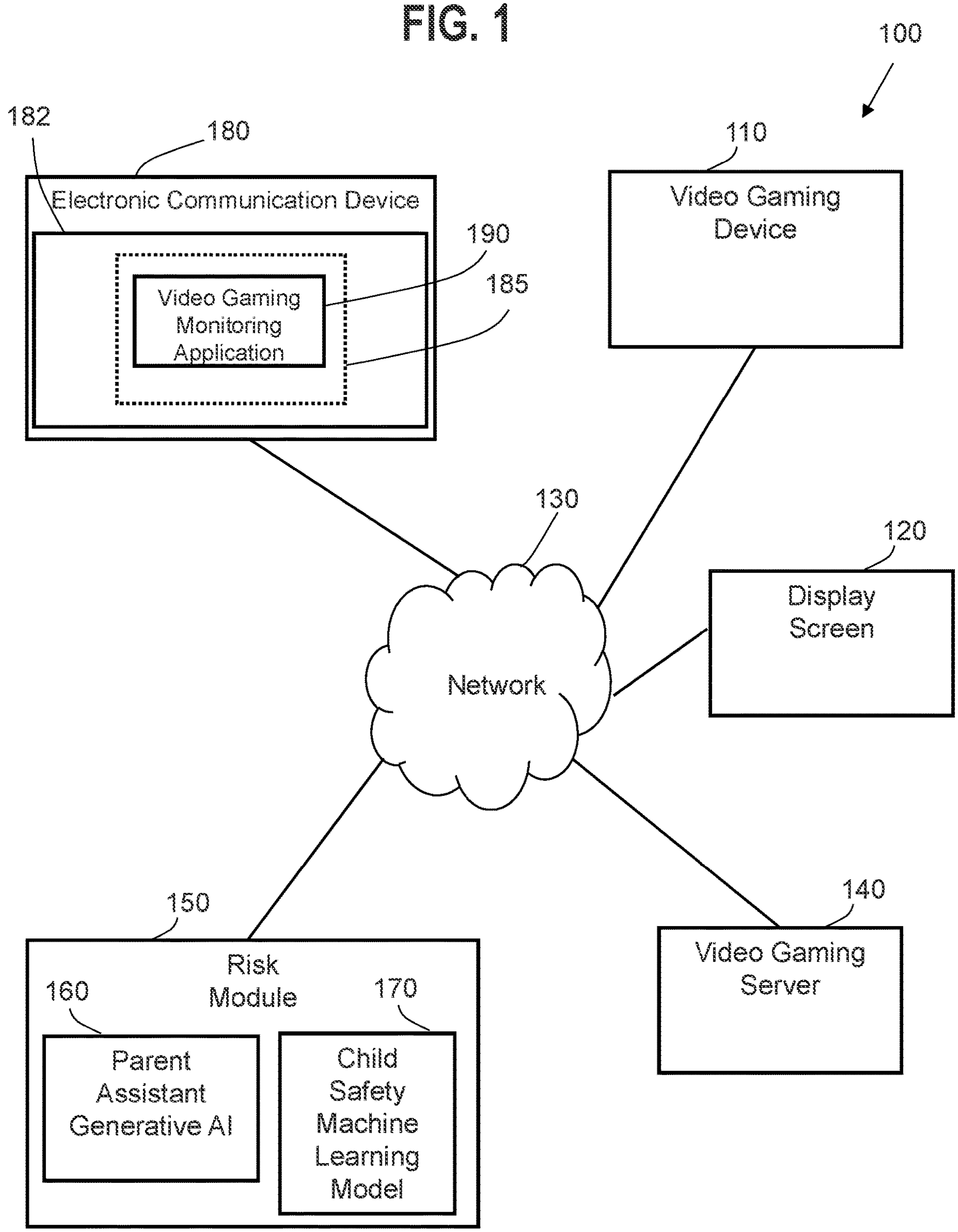

100
182
180
Electronic Communication Device
190
185
Video Gaming Monitoring Application
110
Video Gaming Device
130
Network
120
Display Screen
150
Risk Module
160
Parent Assistant Generative AI
170
Child Safety Machine Learning Model
140
Video Gaming Server

FIG. 2

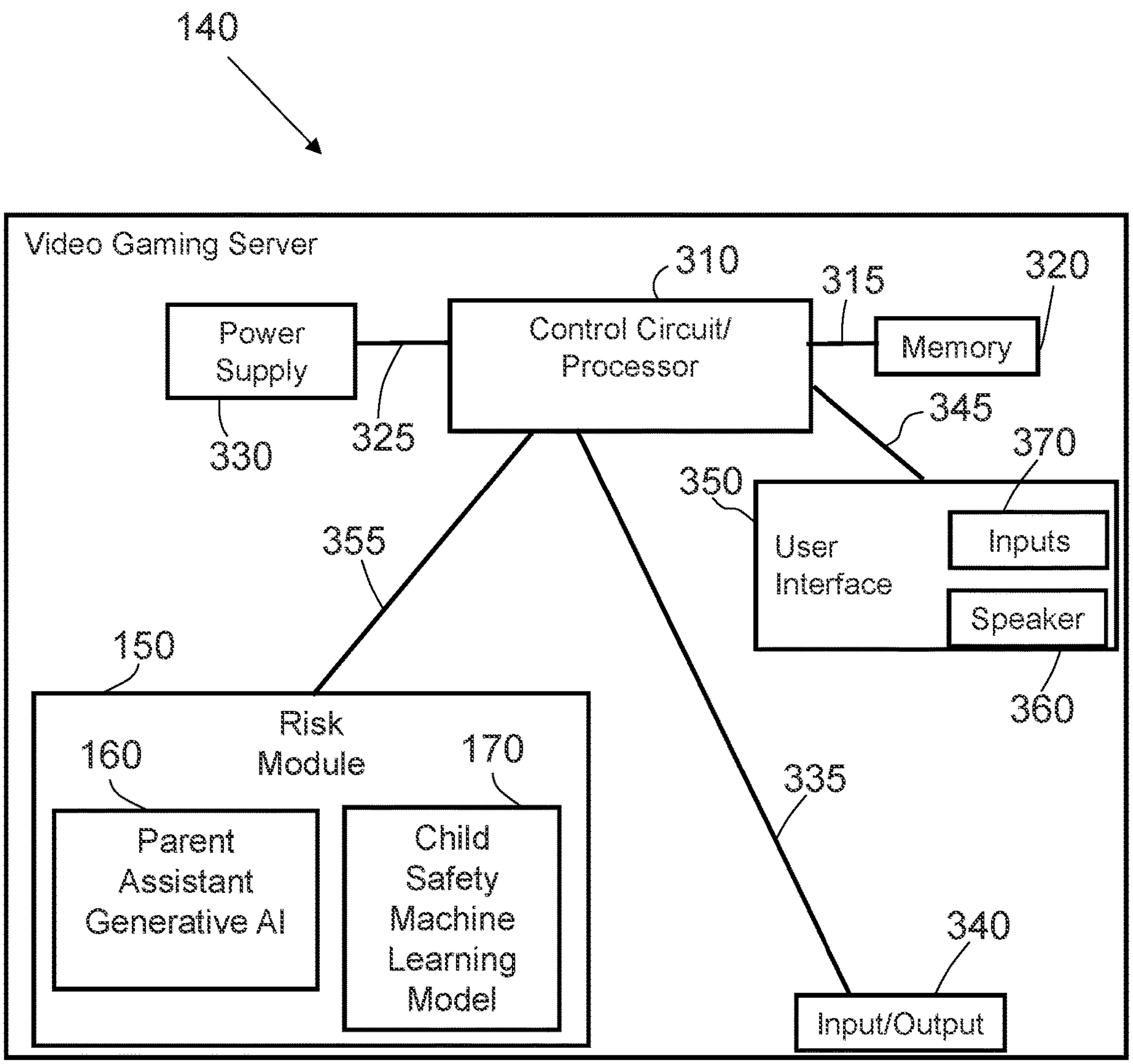

140
Video Gaming Server
310
315
320
Power Supply
Control Circuit/ Processor
Memory
330
325
345
370
350
User Interface
Inputs
Speaker
355
360
150
Risk Module
160
170
Parent Assistant Generative AI
Child Safety Machine Learning Model
335
340
Input/Output

FIG. 3

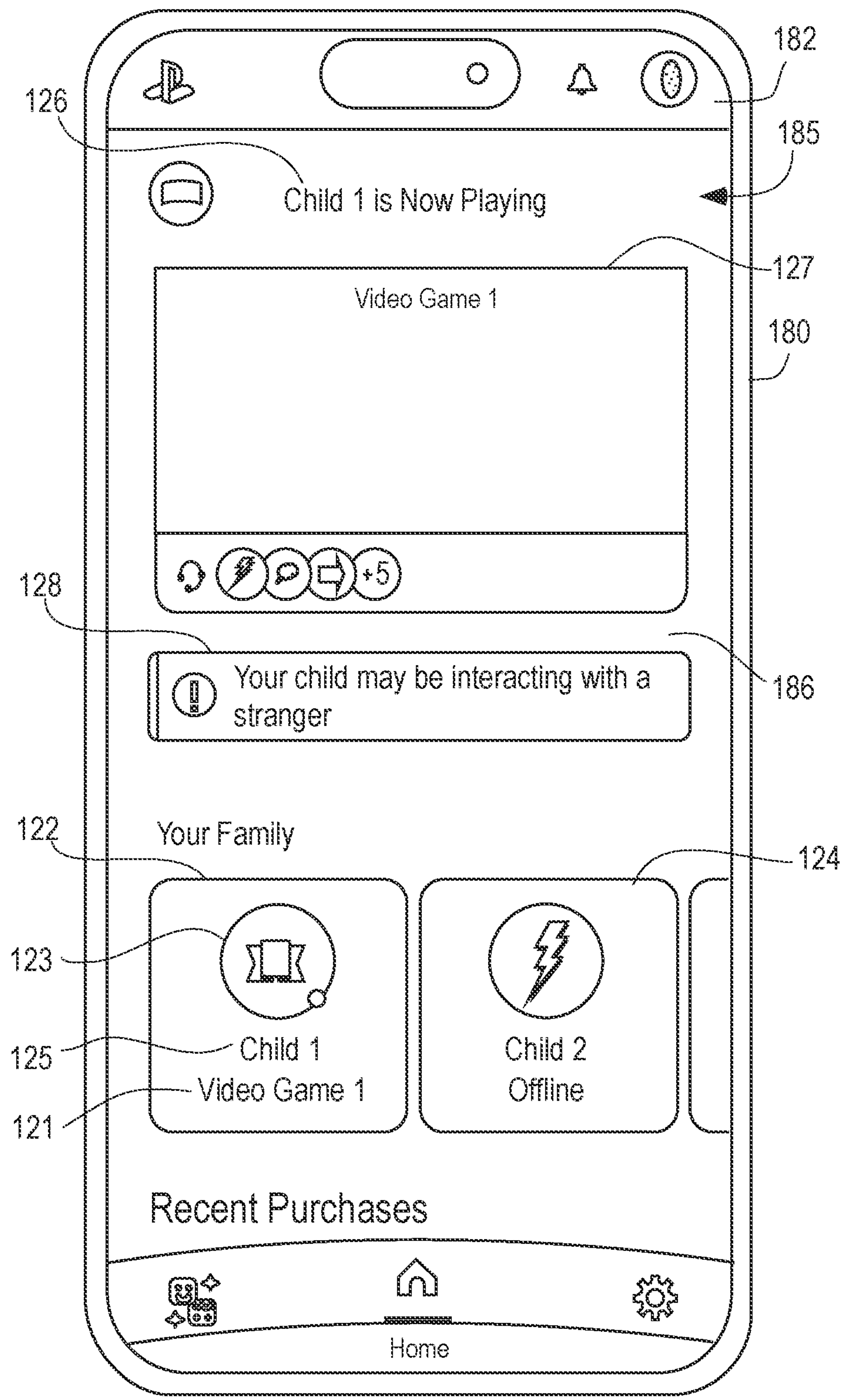
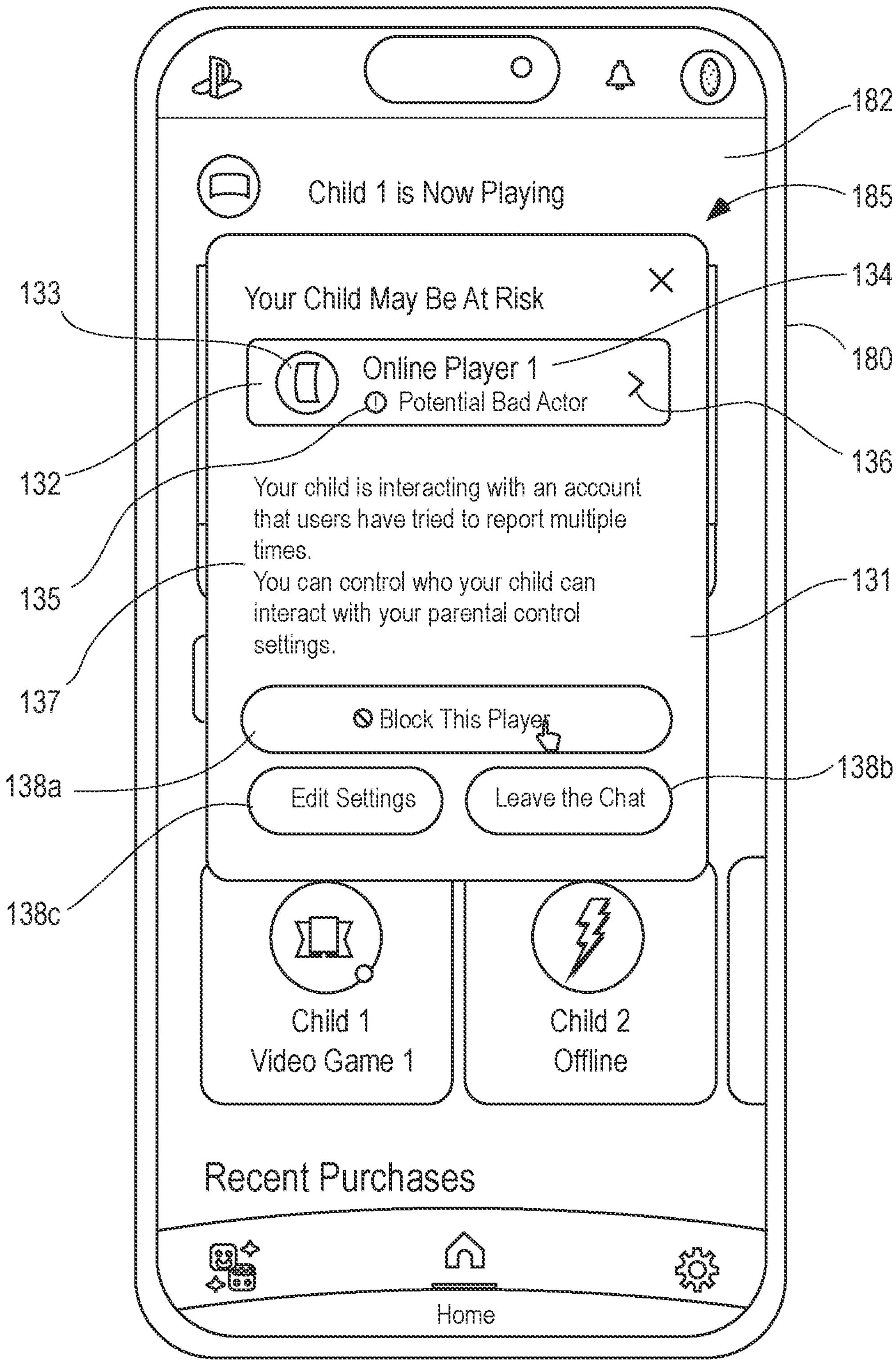

182
126
185
Child 1 is Now Playing
127
Video Game 1
180
+5
128
186
Your child may be interacting with a stranger
Your Family
122
124
123
125
121
Child 1
Video Game 1
Child 2
Offline
Recent Purchases
Home

FIG. 4



182
185
Child 1 is Now Playing
134
Your Child May Be At Risk
133
180
Online Player 1
Potential Bad Actor
132
136
Your child is interacting with an account that users have tried to report multiple times.
You can control who your child can interact with your parental control settings.
135
131
137
Block This Player
138a
138b
Edit Settings
Leave the Chat
138c
Child 1
Video Game 1
Child 2
Offline
Recent Purchases
Home

FIG. 5

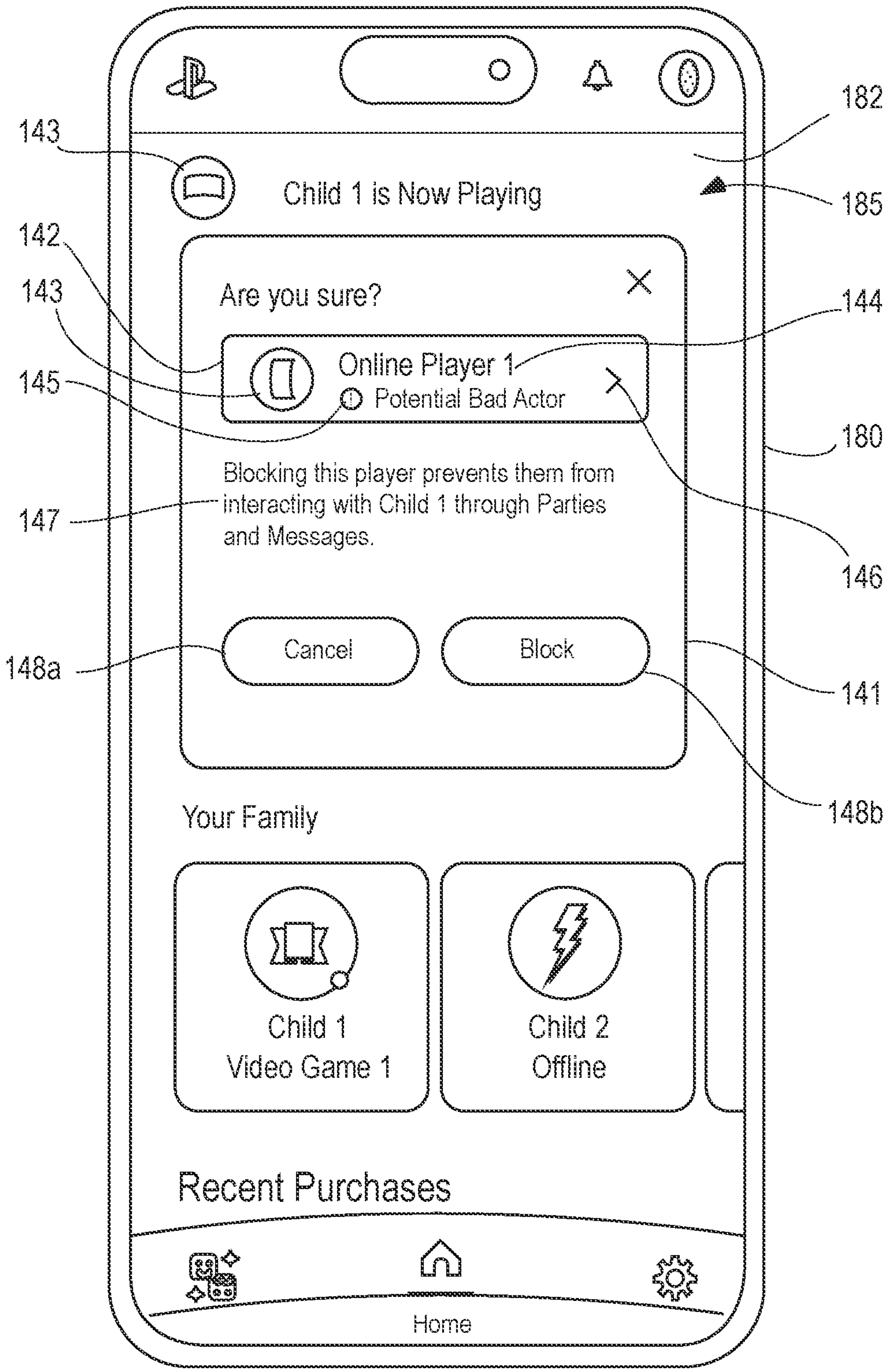

182
143
Child 1 is Now Playing
185
142
Are you sure?
143
144
Online Player 1
145
Potential Bad Actor
180
Blocking this player prevents them from interacting with Child 1 through Parties and Messages.
147
146
148a
Cancel
Block
141
148b
Your Family
Child 1
Video Game 1
Child 2
Offline
Recent Purchases
Home

FIG. 6

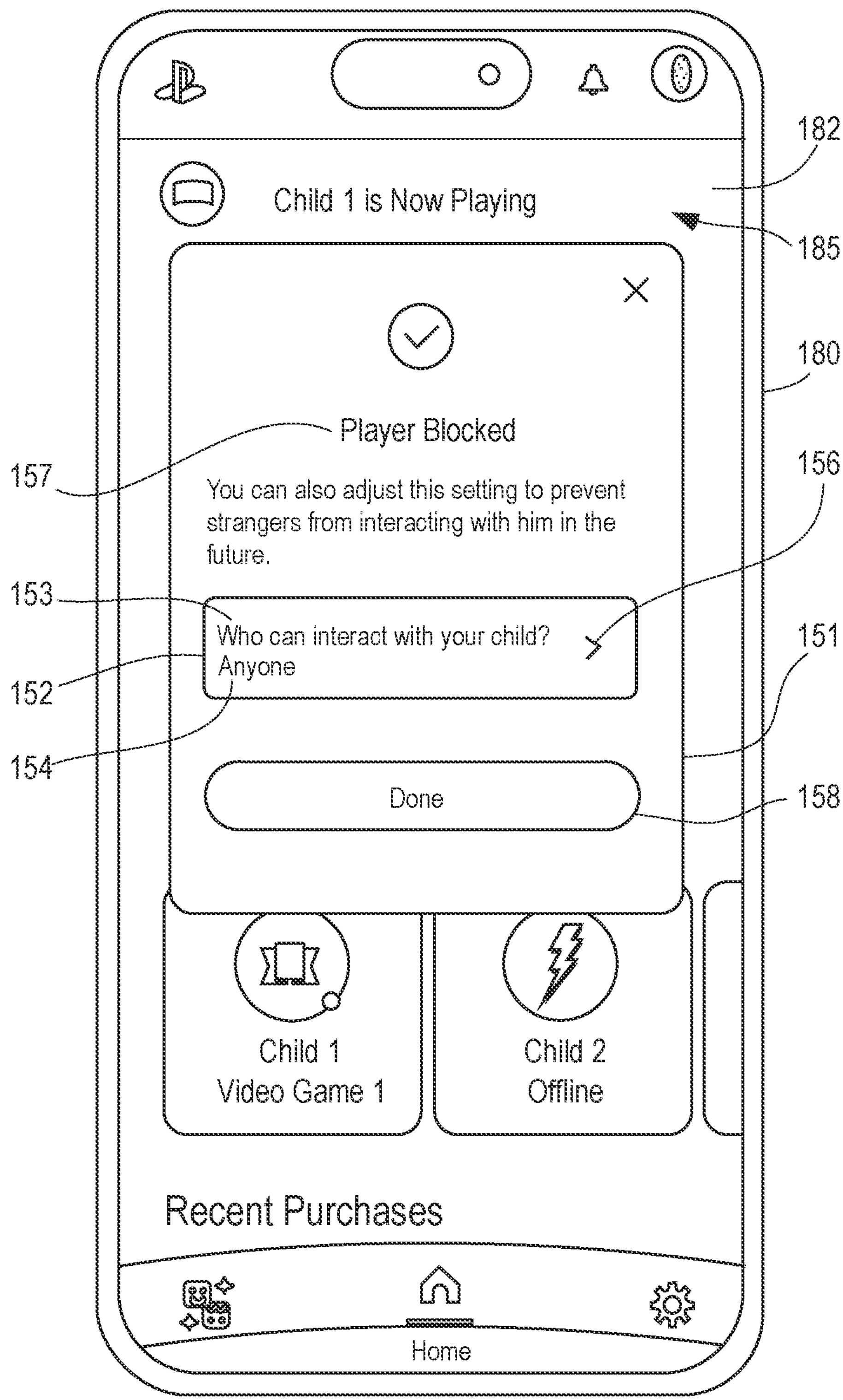

Child 1 is Now Playing
182
185
180
Player Blocked
157
You can also adjust this setting to prevent strangers from interacting with him in the future.
156
153
Who can interact with your child?
Anyone
152
151
154
Done
158
Child 1
Video Game 1
Child 2
Offline
Recent Purchases
Home

FIG. 7

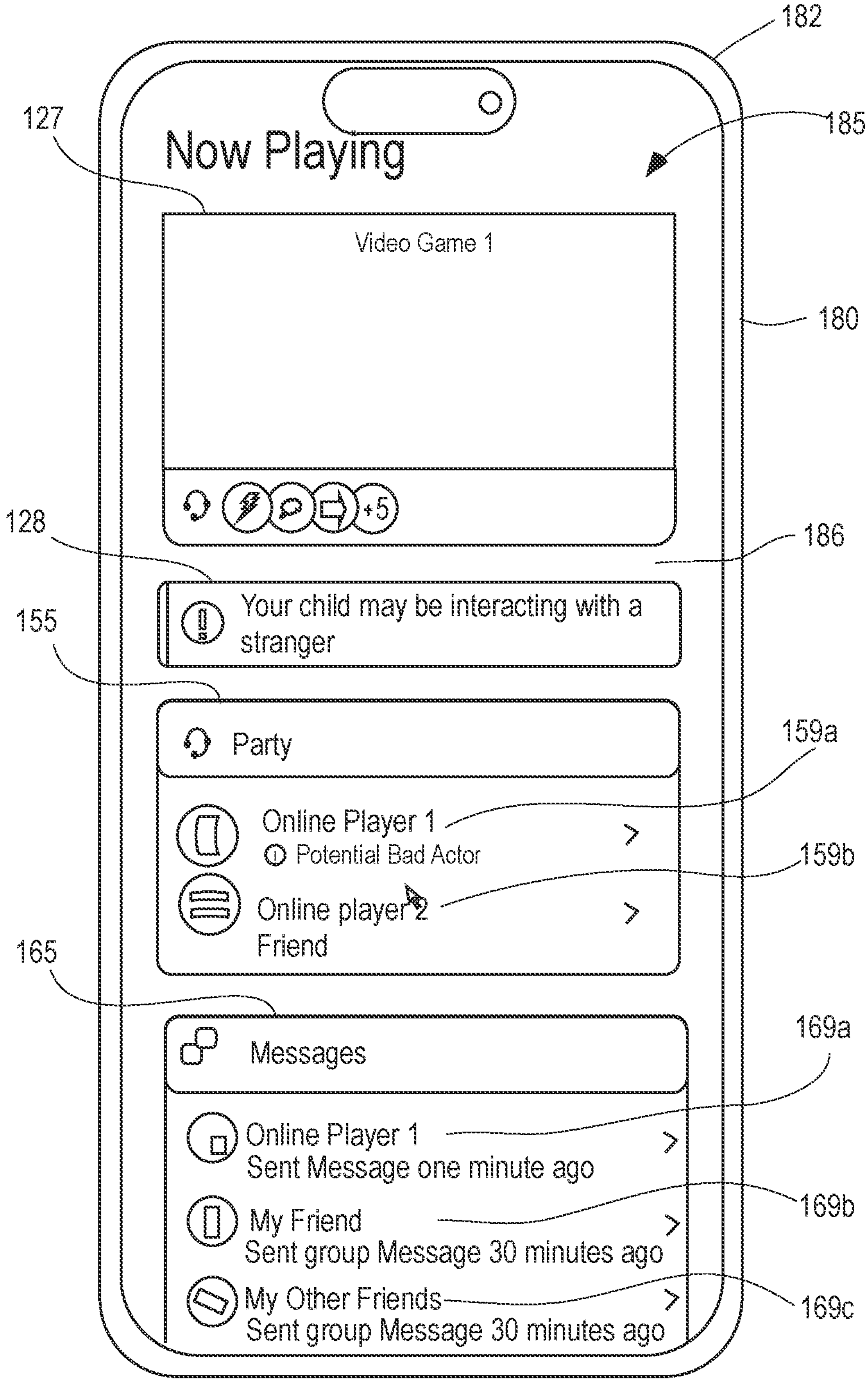

182
185
127
Now Playing
Video Game 1
+5
180
128
186
Your child may be interacting with a stranger
155
Party
159a
Online Player 1
Potential Bad Actor
159b
Online player 2
Friend
165
Messages
169a
Online Player 1
Sent Message one minute ago
169b
My Friend
Sent group Message 30 minutes ago
169c
My Other Friends
Sent group Message 30 minutes ago

FIG. 8

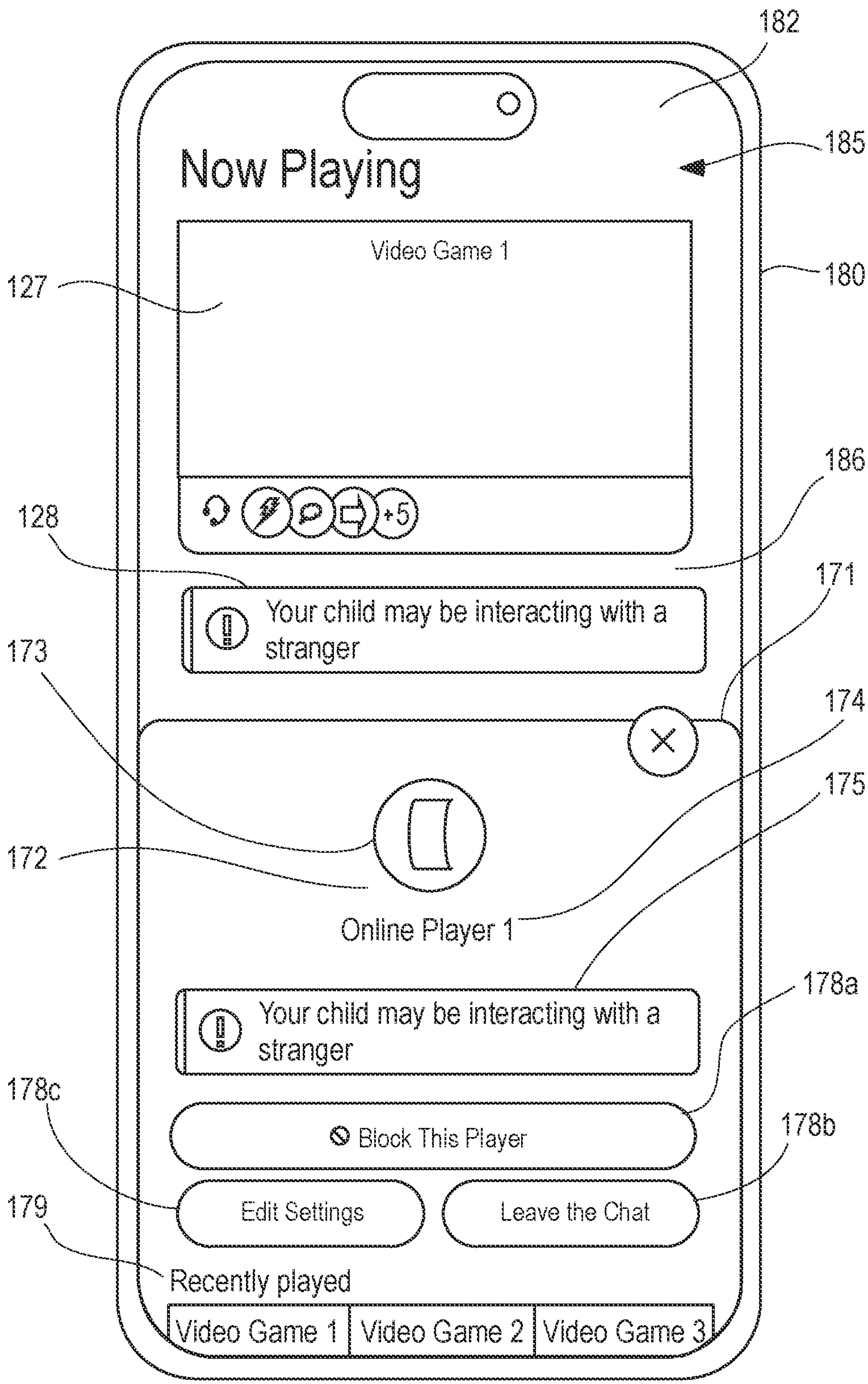

182
185
Now Playing
Video Game 1
127
180
186
128
+5
Your child may be interacting with a stranger
171
173
174
175
172
Online Player 1
Your child may be interacting with a stranger
178a
Block This Player
178c
178b
Edit Settings
Leave the Chat
179
Recently played
Video Game 1
Video Game 2
Video Game 3

FIG. 9

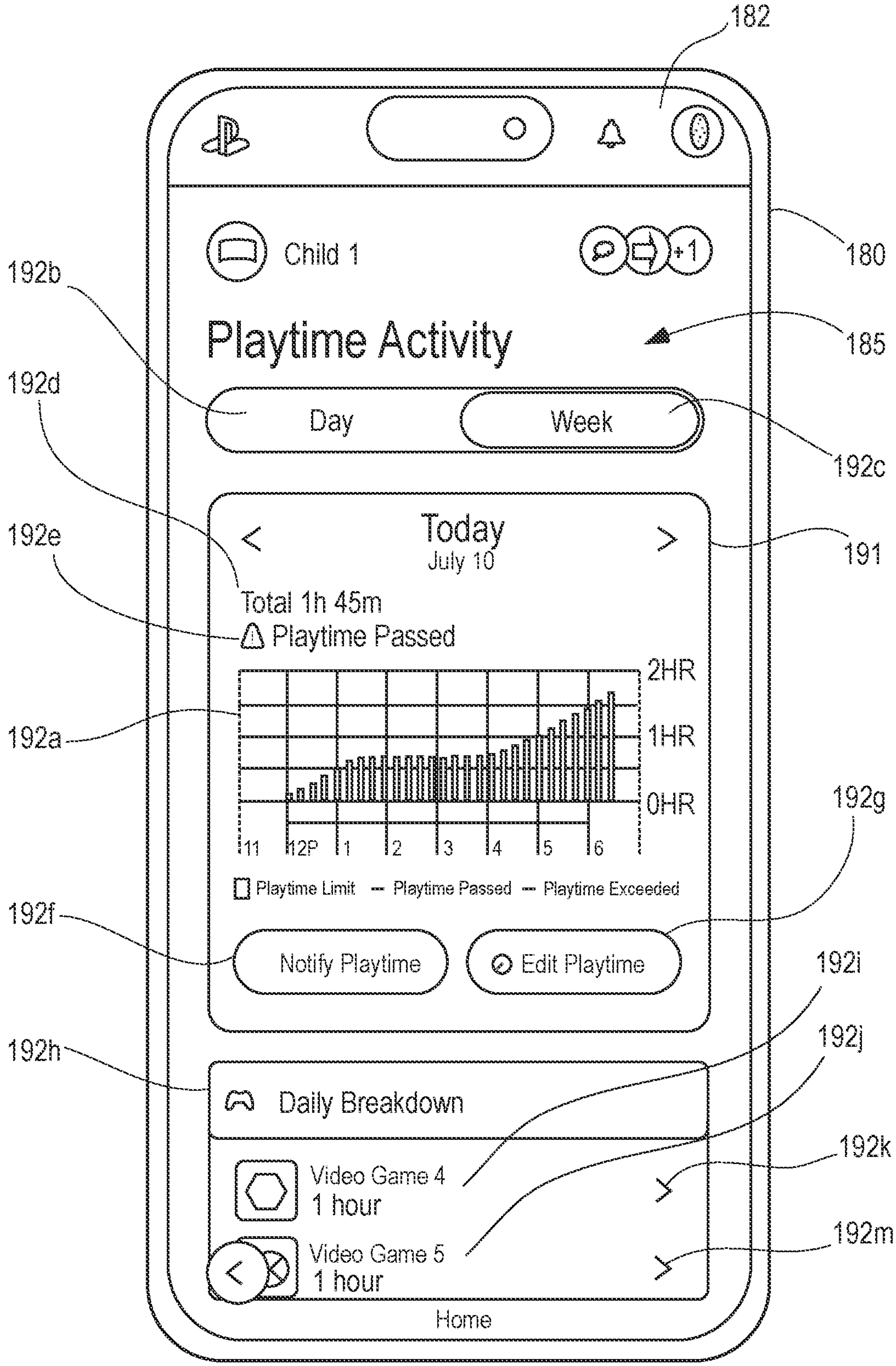

182
180
Child 1
+1
185
Playtime Activity
192b
Day
Week
192c
192d
Today
July 10
191
192e
Total 1h 45m
Playtime Passed
2HR
1HR
0HR
192a
11
12P
1
2
3
4
5
6
Playtime Limit
Playtime Passed
Playtime Exceeded
192g
192f
Notify Playtime
Edit Playtime
192i
192h
Daily Breakdown
192j
Video Game 4
1 hour
192k
Video Game 5
1 hour
192m
Home

FIG. 10

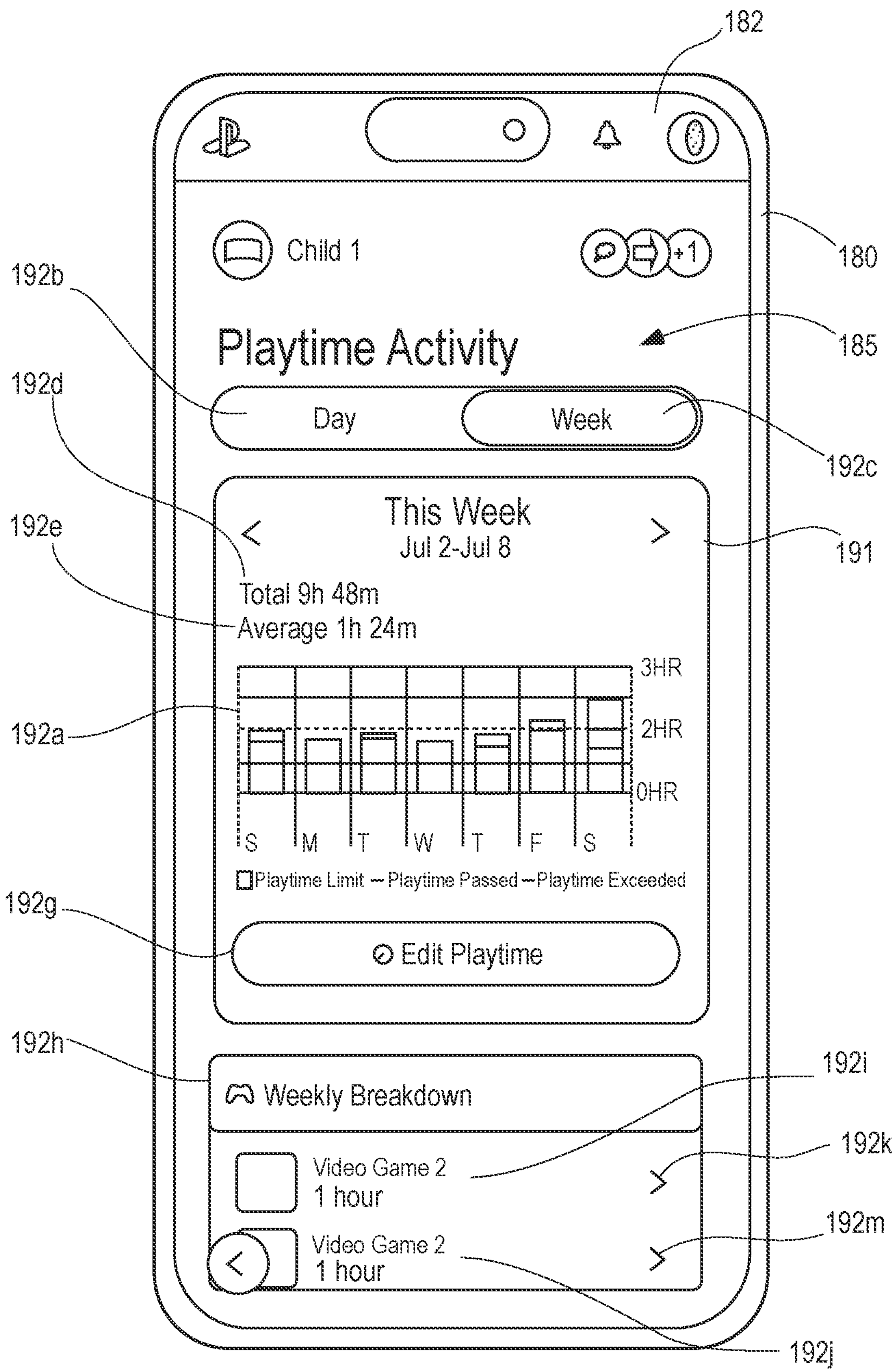
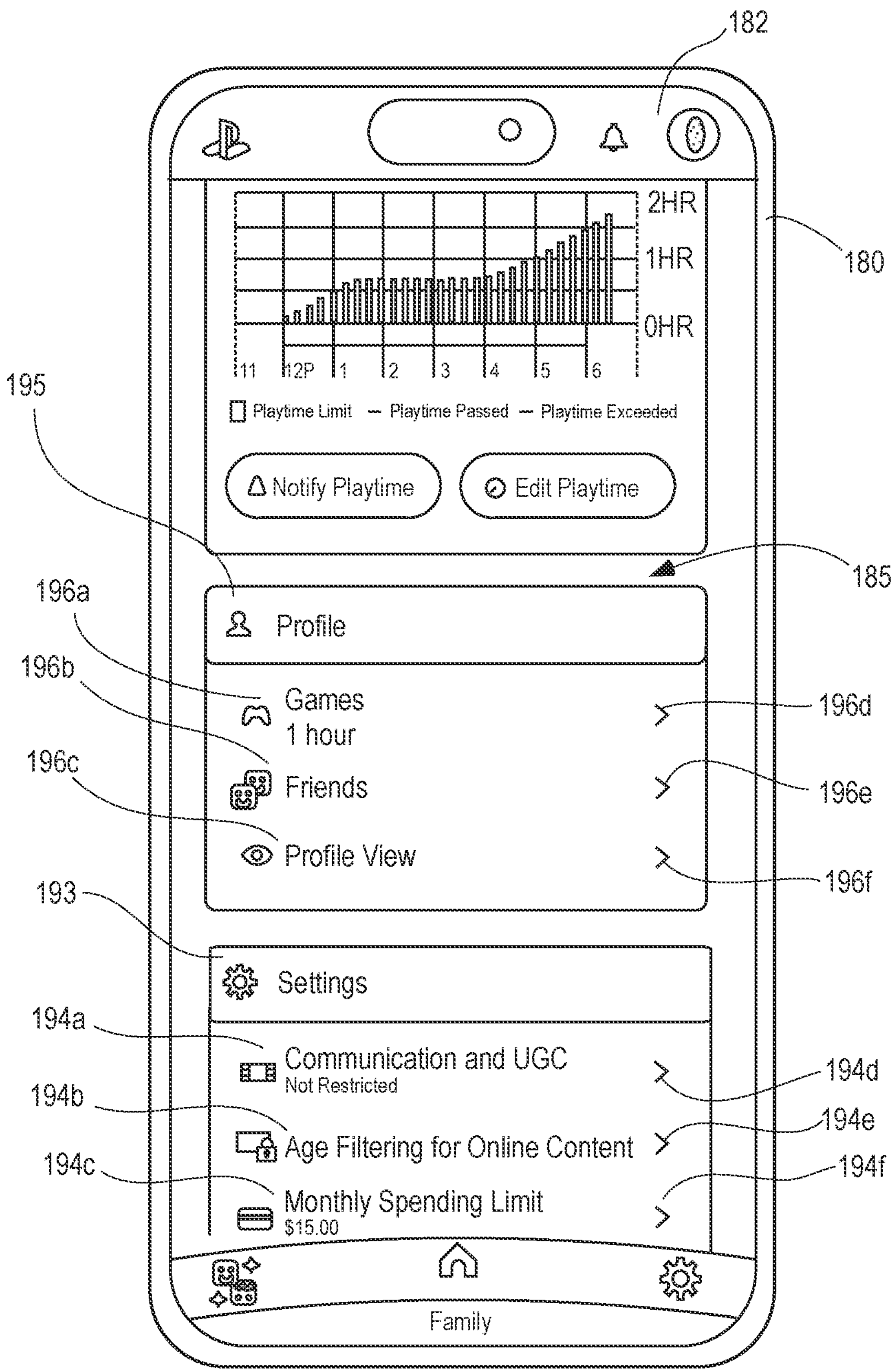

182
180
185
Child 1
+1
Playtime Activity
Day
Week
This Week
Jul 2-Jul 8
Total 9h 48m
Average 1h 24m
3HR
2HR
0HR
S M T W T F S
Playtime Limit
Playtime Passed
Playtime Exceeded
Edit Playtime
Weekly Breakdown
Video Game 2
1 hour
Video Game 2
1 hour
192a
192b
192c
192d
192e
192g
192h
192i
192j
192k
192m
191

FIG. 11



182
180
2HR
1HR
0HR
11
12P
1
2
3
4
5
6
Playtime Limit
Playtime Passed
Playtime Exceeded
Notify Playtime
Edit Playtime
195
185
Profile
196a
196b
196c
Games
1 hour
Friends
Profile View
196d
196e
196f
193
Settings
194a
194b
194c
Communication and UGC
Not Restricted
Age Filtering for Online Content
Monthly Spending Limit
$15.00
194d
194e
194f
Family

FIG. 12

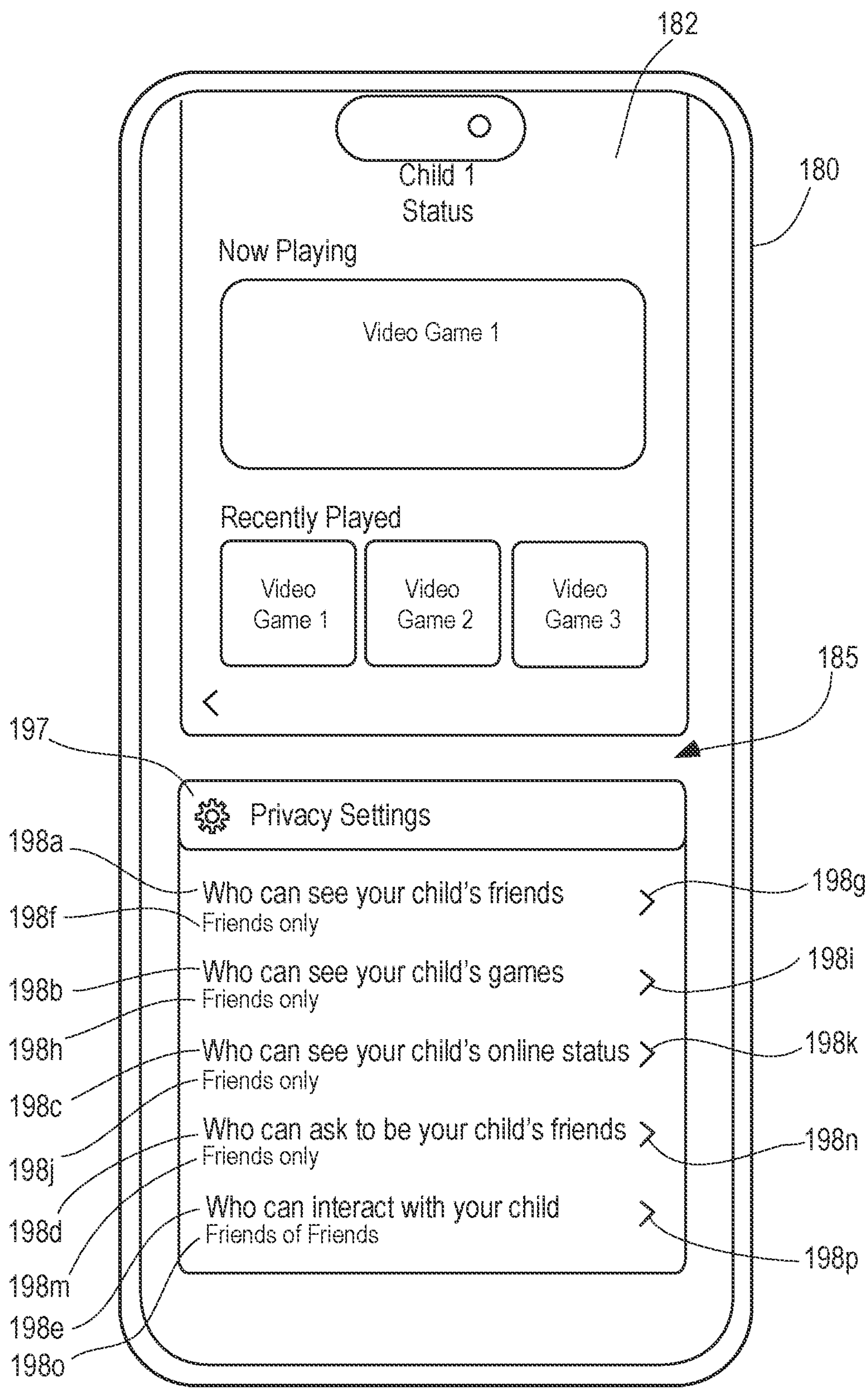

182
180
Child 1
Status
Now Playing
Video Game 1
Recently Played
Video Game 1
Video Game 2
Video Game 3
185
197
Privacy Settings
198a
Who can see your child's friends
198f
Friends only
198g
198b
Who can see your child's games
198h
Friends only
198i
198c
Who can see your child's online status
198j
Friends only
198k
198d
Who can ask to be your child's friends
198m
Friends only
198n
198e
Who can interact with your child
198o
Friends of Friends
198p

FIG. 13

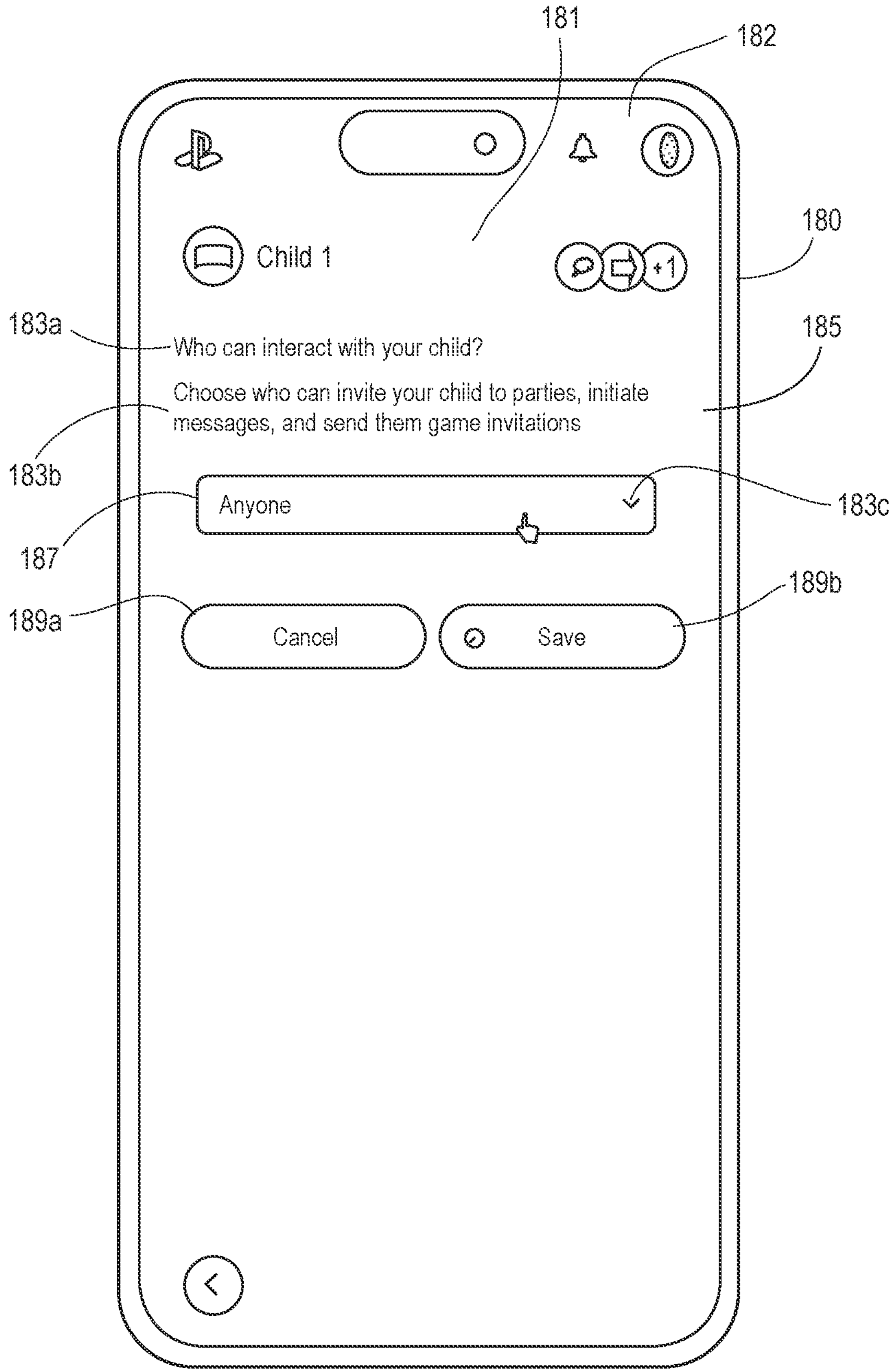
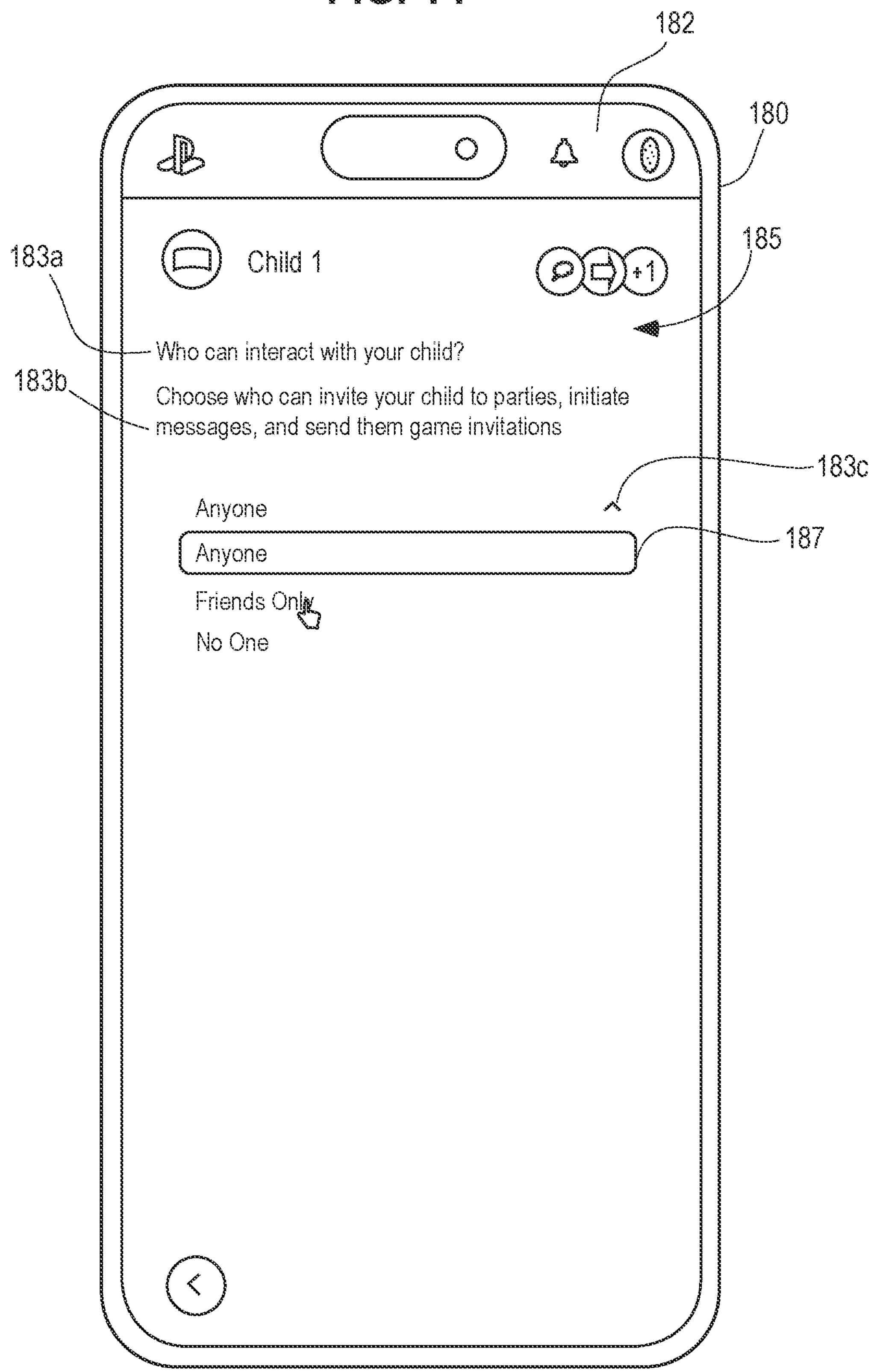

181
182
180
Child 1
+1
183a
Who can interact with your child?
185
Choose who can invite your child to parties, initiate messages, and send them game invitations
183b
Anyone
183c
187
189b
189a
Cancel
Save

FIG. 14



182
180
Child 1
+1
185
183a
Who can interact with your child?
183b
Choose who can invite your child to parties, initiate messages, and send them game invitations
183c
Anyone
Anyone
187
Friends Only
No One

FIG. 15

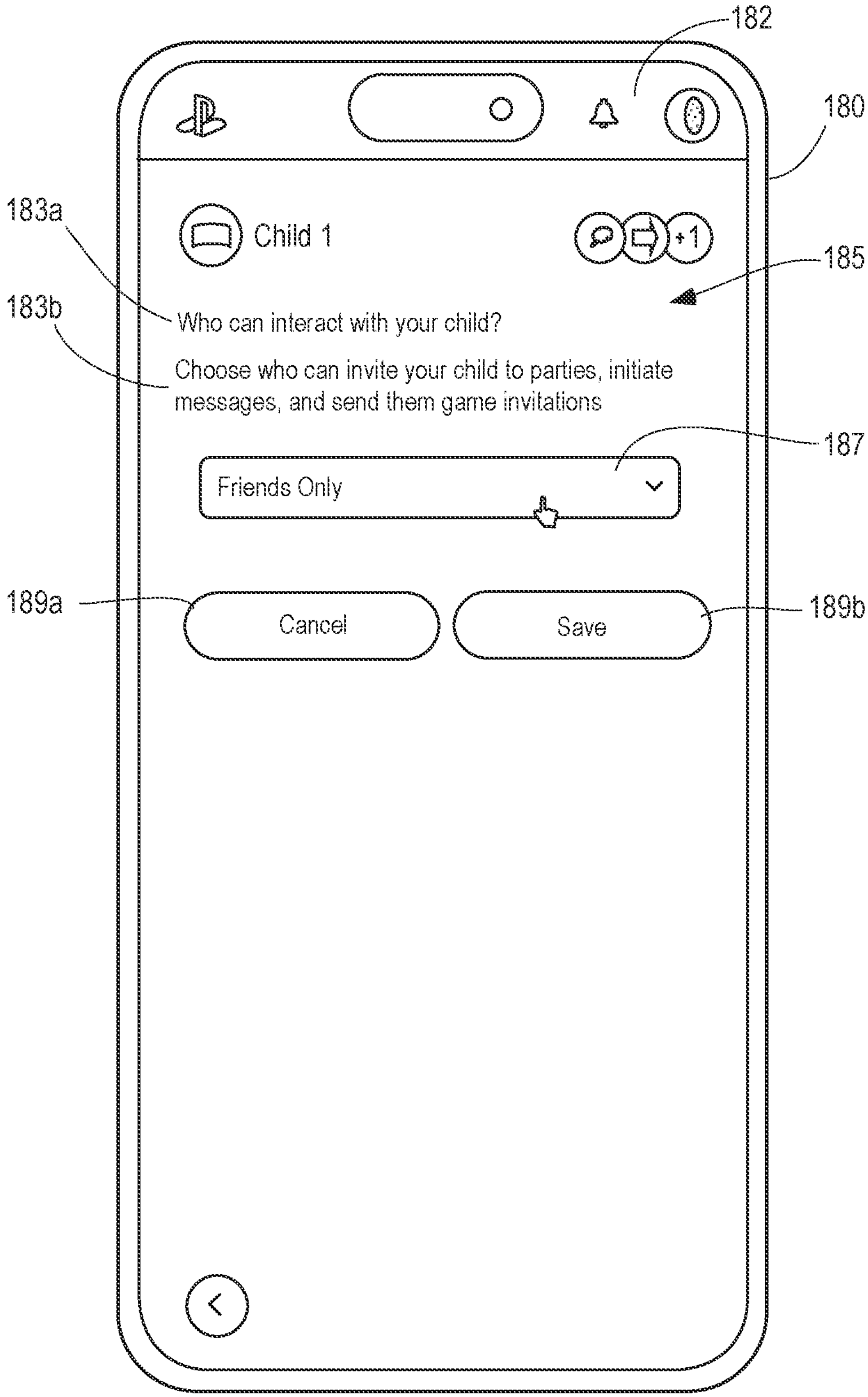

182
180
183a
Child 1
+1
185
183b
Who can interact with your child?
Choose who can invite your child to parties, initiate messages, and send them game invitations
187
Friends Only
189a
Cancel
Save
189b

FIG. 16

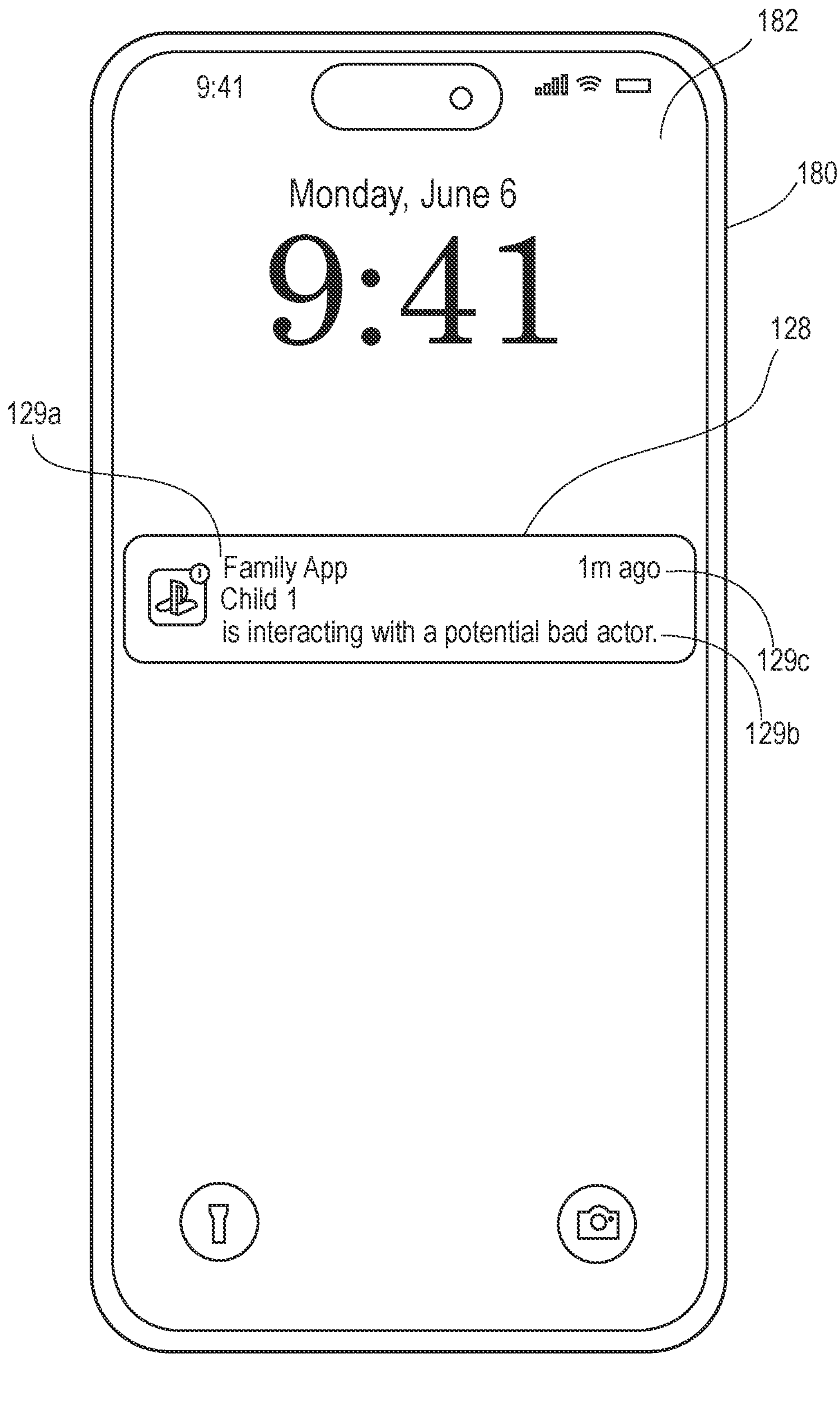
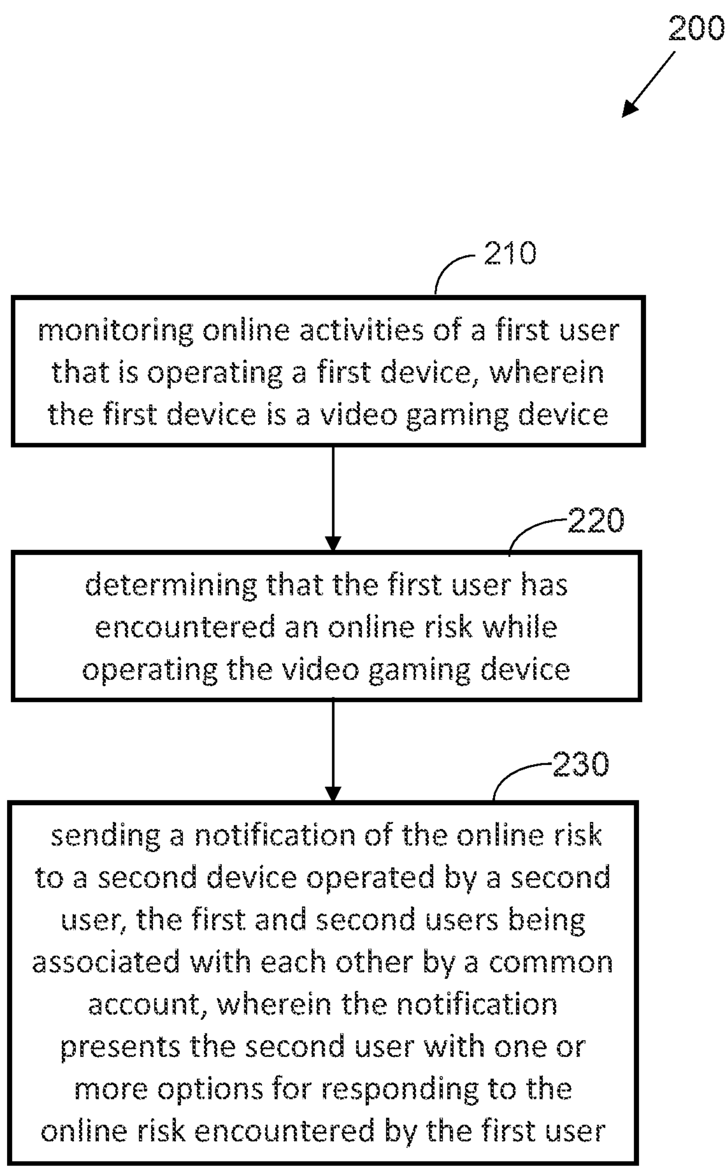

182
180
9:41
Monday, June 6
9:41
128
129a
Family App
1m ago
Child 1
is interacting with a potential bad actor.
129c
129b

FIG. 17


200
210
monitoring online activities of a first user that is operating a first device, wherein the first device is a video gaming device
220
determining that the first user has encountered an online risk while operating the video gaming device
230
sending a notification of the online risk to a second device operated by a second user, the first and second users being associated with each other by a common account, wherein the notification presents the second user with one or more options for responding to the online risk encountered by the first user

ON-LINE GAMING MONITORING SYSTEMS AND USER INTERFACES

TECHNICAL FIELD

[0001] This invention relates generally to monitoring systems, and, more specifically, to systems and user interfaces for monitoring online video game play of one or more users.

BACKGROUND

[0002] Online gaming has become increasingly popular among children, providing them with opportunities for entertainment, social interaction, and skill development. However, it is essential to be aware of the potential risks associated with online gaming. One significant risk is exposure to inappropriate content, such as violence or explicit language, which can negatively impact a child's behavior and well-being. Another concern is online predators who may exploit children's vulnerabilities. Additionally, excessive gaming can affect academic performance, physical health, and social interactions. It is crucial for parents to establish clear guidelines, monitor online activities, and promote a healthy balance between gaming and other activities to ensure a safe and positive gaming experience for children.

SUMMARY

[0003] In some embodiments, a method includes monitoring online activities of a first user that is operating a first device, wherein the first device is a video gaming device, determining that the first user has encountered an online risk while operating the video gaming device, and sending a notification of the online risk to a second device operated by a second user, the first and second users being associated with each other by a common account. The notification presents the second user with one or more options for responding to the online risk encountered by the first user.

[0004] In one aspect, the online risk comprises an interaction of the first user with a third user, the interaction including at least one of a message, an image, a video, or an audio.

[0005] In some implementations, the determining that the first user has encountered an online risk includes detecting that an interaction between a third user and the first user violates one or more permissions or restrictions pre-set by the second user for the first user.

[0006] In certain embodiments, the determining that the first user has encountered an online risk includes feeding information relating to online activity of the first user while the first user is operating the video gaming device into an artificial intelligence model trained to recognize the online risk to the first user.

[0007] In some aspects, the one or more options presented to the second user for responding to the online risk encountered by the first user include one or more of: blocking a third user, reporting the third user, flagging the third user, sending a message to the third user, viewing online activity of the first user in real time, modifying the online activity of the first user in real time, sending a message to the first user, viewing recorded online activity of the first user, monitoring for more information regarding the online risk, modifying one or more permissions or restrictions for the online activity of the first user, or ignoring the online risk.

[0008] In one embodiment, the notification further includes a description of the online risk and/or content associated with the online risk.

[0009] In some implementations, the notification is transmitted from a video gaming server to a mobile application installed on the second device, and the notification is displayed within a graphical interface generated by the mobile application.

[0010] In some embodiments, a system includes a video gaming server configured to communicate with a video gaming device of a first user and a risk module coupled to the video gaming server. The risk module is configured to: monitor online activities of the first user when the first user operates the video gaming device; determine that the first user has encountered an online risk while operating the video gaming device; and cause a notification of the online risk to be sent to a second device operated by a second user, the first and second users being associated with each other by a common account. The notification presents the second user with one or more options for responding to the online risk encountered by the first user.

[0011] In certain embodiments, a non-transitory computer readable storage medium stores one or more computer programs which, when executed, execute steps comprising: in response to receipt of a communication indicating that a first user operating a video gaming device has encountered an online risk, displaying a notification of the online risk on an electronic device associated with a second user, the first and second users being associated with each other by a common account; displaying on the electronic device associated with the second user one or more options for responding to the online risk encountered by the first user; and sending by the electronic device associated with the second user the one or more options selected by the second user to a video gaming server that at least partially controls online activity of the first user while operating the video gaming device.

[0012] A better understanding of the features and advantages of various embodiments of the present invention will be obtained by reference to the following detailed description and accompanying drawings which set forth illustrative embodiments of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] Disclosed herein are embodiments of systems, user interfaces, and methods pertaining to monitoring and/or managing online video game play of one or more users. This description includes drawings, wherein:

[0014] FIG. **1** is a diagram of a system of monitoring online video game play of one or more users in accordance with some embodiments;

[0015] FIG. **2** is a functional diagram of an exemplary video gaming server usable with the system of FIG. **1** in accordance with some embodiments;

[0016] FIG. **3** depicts an electronic communication device displaying a graphical user interface generated by an application executable on the electronic communication device and configured to permit a user of the electronic communication device to monitor online video gaming activities of one or more users linked to the user of the electronic communication device by a common account in accordance with some embodiments;

[0017] FIG. **4** depicts the electronic communication device of FIG. **3**, displaying an interactive menu of the graphical user interface generated by the application in response to a user interaction with the graphical user interface of FIG. **3** in accordance with some embodiments;

[0018] FIG. **5** depicts the electronic communication device of FIG. **3** displaying an interactive menu of the graphical user interface generated by the application in response to a user interaction with the graphical user interface of FIG. **4** in accordance with some embodiments;

[0019] FIG. **6** depicts the electronic communication device of FIG. **3** displaying an interactive menu of the graphical user interface generated by the application in response to a user interaction with the graphical user interface of FIG. **5** in accordance with some embodiments;

[0020] FIG. **7** depicts an electronic communication device displaying a graphical user interface generated by an application executable on the electronic communication device and configured to permit a user of the electronic communication device to monitor online video gaming activities of one or more users linked to the user of the electronic communication device by a common account in accordance with some embodiments;

[0021] FIG. **8** depicts the electronic communication device of FIG. **7** displaying an interactive menu of the graphical user interface generated by the application in response to a user interaction with the graphical user interface of FIG. **7** in accordance with some embodiments;

[0022] FIG. **9** depicts an electronic communication device displaying a graphical user interface generated by an application executable on the electronic communication device and configured to permit a user of the electronic communication device to monitor online video gaming activities of one or more users linked to the user of the electronic communication device by a common account in accordance with some embodiments;

[0023] FIG. **10** depicts the electronic communication device of FIG. **9** displaying an interactive menu of the graphical user interface generated by the application in response to a user interaction with the graphical user interface of FIG. **9** in accordance with some embodiments;

[0024] FIG. **11** depicts the electronic communication device of FIG. **9** displaying an interactive menu of the graphical user interface generated by the application in response to a user interaction with the graphical user interface of FIG. **10** in accordance with some embodiments;

[0025] FIG. **12** depicts an electronic communication device displaying a graphical user interface generated by an application executable on the electronic communication device and configured to permit a user of the electronic communication device to monitor online video gaming activities of one or more users linked to the user of the electronic communication device by a common account in accordance with some embodiments;

[0026] FIG. **13** depicts the electronic communication device of FIG. **12** displaying an interactive menu of the graphical user interface generated by the application in response to a user interaction with the graphical user interface of FIG. **12** in accordance with some embodiments;

[0027] FIG. **14** depicts the electronic communication device of FIG. **12** displaying an interactive menu of the graphical user interface generated by the application in response to a user interaction with the graphical user interface of FIG. **13** in accordance with some embodiments;

[0028] FIG. **15** depicts the electronic communication device of FIG. **12** displaying an interactive menu of the graphical user interface generated by the application in response to a user interaction with the graphical user interface of FIG. **14** in accordance with some embodiments;

[0029] FIG. **16** depicts an electronic communication device displaying a notification generated by an application executable on the electronic communication device and configured to permit a user of the electronic communication device to monitor online video gaming activities of one or more users linked to the user of the electronic communication device by a common account in accordance with some embodiments;

[0030] FIG. **17** is a flow diagram representative of a method of monitoring online video game play of one or more users in accordance with some embodiments.

[0031] Elements in the figures are illustrated for simplicity and clarity and have not been drawn to scale. For example, the dimensions and/or relative positioning of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of various embodiments of the present invention. Also, common but well-understood elements that are useful or necessary in a commercially feasible embodiment are often not depicted in order to facilitate a less obstructed view of these various embodiments of the present invention. Certain actions and/or steps may be described or depicted in a particular order of occurrence while those skilled in the art will understand that such specificity with respect to sequence is not actually required. The terms and expressions used herein have the ordinary technical meaning as is accorded to such terms and expressions by persons skilled in the technical field as set forth above except where different specific meanings have otherwise been set forth herein.

DETAILED DESCRIPTION

[0032] The following description is not to be taken in a limiting sense, but is made merely for the purpose of describing the general principles of exemplary embodiments. Reference throughout this specification to "one embodiment," "an embodiment," or similar language means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the present invention. Thus, appearances of the phrases "in one embodiment," "in an embodiment," and similar language throughout this specification may, but do not necessarily, all refer to the same embodiment.

[0033] Generally speaking, pursuant to various embodiments, systems and methods described herein include a video gaming server configured to communicate with a video gaming device of a first user. A risk module is coupled to the video gaming server. The risk module is configured to: monitor online activities of the first user when the first user operates the video gaming device; determine that the first user has encountered an online risk while operating the video gaming device; and cause a notification of the online risk to be sent to a second device operated by a second user, the first and second users being associated with each other by a common account. The notification presents the second user with one or more options for responding to the online risk encountered by the first user.

[0034] FIG. **1** shows an exemplary embodiment of a system **100** for monitoring online video game play of one or more users. The exemplary system **100** shown in FIG. **1** includes a video gaming device **110** (which may be one or more computing devices as pointed out below) operatively coupled/connected to a display screen **120** and configured to communicate over a network **130** (e.g., a wired or wireless connection) with the display screen **120**. The exemplary system **100** further includes a video gaming server **140**, which permits gamers to use their respective video gaming devices **110** to play video games online together over the network **130**, and which enables users (e.g., parents, guardians, etc.) to monitor this online gaming, restrict exposure of the gamers to online risks, and/or address/eliminate any active online risks.

[0035] The exemplary network **130** depicted in FIG. **1** may be a wide-area network (WAN), a local area network (LAN), a personal area network (PAN), a wireless local area network (WLAN), Wi-Fi, Zigbee, Bluetooth (e.g., Bluetooth Low Energy (BLE) network), or any other internet or intranet network, or combinations of such networks. Generally, communication between various electronic devices of system **100** may take place over hard-wired, wireless, cellular, Wi-Fi or Bluetooth networked components or the like. In some embodiments, one or more electronic devices of system **100** may include cloud-based features or services, such as cloud-based memory storage, cloud-based machine learning models/neural networks, etc.

[0036] The video gaming device **110** may be a stationary or portable electronic device, for example, a stationary gaming console, a portable gaming console, a desktop computer, a laptop computer, a tablet, a mobile phone, a single server or a series of communicatively connected servers, or any other electronic device including a control circuit that includes a programmable processor and may be coupled/connected to the display screen **120**. In some embodiments, the video gaming device **110** is configured for running video games thereon (e.g., from a disc inserted into the video gaming device **110**, from an onboard memory of the video gaming device **110**, from a remote server/host, etc.). The video gaming device **110** may be configured for data entry and processing and for communication with other devices of the system **100** via the network **130**.

[0037] The exemplary system **100** further includes a risk module **150**, which may be a stand-alone physical device (or multiple devices) as shown in FIG. **1**, or may be incorporated (e.g., embedded) into the video gaming device **110** and/or the video gaming server **140** (an embodiment of an exemplary video gaming server **140** incorporating the risk module **150** is shown in FIG. **2**). In the illustrated embodiment, the risk module **150** includes or is coupled to a generative artificial intelligence (AI) component **160**, as well as a child safety machine learning model **170**.

[0038] Generally, the term generative AI component **160** refers to a module or algorithm within an AI system that is capable of creating new content (e.g., text, images, music, etc.) based on patterns and examples it has been trained on. Unlike traditional AI models that rely on predefined rules or templates, generative AI models have the ability to generate novel and creative outputs that mimic human-like behavior. In some aspects, the generative AI component **160** may be built using deep learning techniques, such as recurrent neural networks (RNNs) or generative adversarial networks (GANs), which enable the generative AI component **160** to learn from large datasets and generate new content by leveraging the patterns and structures it has learned.

[0039] Generally, the term child safety machine learning model **170** refers to an algorithm or system that has been trained to detect and address potential risks or threats to the safety of children. A child safety machine learning model **170** may be designed to analyze various types of data, such as text (e.g., such as would be present in online messages), images, voice (e.g., words spoken/received via a microphone/headset while playing online) or online behavior, to identify content or activities that may be harmful or inappropriate for children. Without wishing to be limited by theory, the child safety machine learning model **170** can be utilized in different contexts, such as social media platforms, online gaming environments, or content filtering systems to help identify and filter out content that contains explicit or violent material, cyberbullying, or other forms of online behavior or interactions that may pose risks to children.

[0040] In some aspects, the child safety machine learning model **170** is trained using large datasets that include examples of both safe and unsafe content, which allows the child safety machine learning model **170** to learn to recognize patterns, features, and context that are indicative of potential risks to children. In certain aspects, by leveraging machine learning techniques like deep neural networks, decision trees, or support vector machines, the child safety machine learning model **170** can make predictions and classify content as safe or potentially harmful.

[0041] In the embodiment illustrated in FIG. **1**, the system **100** further includes an electronic communication device **180**, which may be, for example, a smart phone, tablet, laptop computer, desktop computer, etc.). The electronic communication device **180** includes a screen **182** that may display a graphical user interface **185**, which in turn may display a video gaming monitoring application **190**. In some embodiments, the video gaming monitoring application **190** is configured to provide one or more primary users (e.g., adults, parents, guardians, etc.) having an account set up within the video gaming monitoring application **190** with the ability to monitor the video game-related activities (e.g., online gaming) of one or more secondary users (e.g., children, etc.) associated with the account set up by one or more of the primary users.

[0042] In some embodiments, the video gaming monitoring application **190** can be a dedicated application (e.g., an application specific to the electronic communication device **180**) or a general application that that can provide or support video game play monitoring functions. In some embodiments, the video gaming monitoring application **190** can be a browser-based application that is native to an operating system of a remote computer or server or downloaded to and installed on the electronic communication device **180**. In some embodiments, the video gaming monitoring application **190** can provide a link to a browser application on a remote computer or server to cause the browser application to display the user interface **185** on the electronic communication device **180**.

[0043] In some embodiments, a primary user is permitted to download the video gaming monitoring application **190** and set up an account for use in monitoring and/or managing online video game play of one or more secondary users. In one aspect, when a primary user initially sets up a "family" account via the video gaming device **110** and/or the video gaming server **140** for monitoring and/or managing online video game play of one or more users associated with this account, the video gaming monitoring application **190** is configured to permit the primary user to generate an account profile, which may include a listing including all primary and secondary users associated with the account, personal information (e.g., name, address, age, etc.) of all users associated with the account, various options for setting up various permissions/restrictions for the secondary users, payment methods (e.g., credit card info), and/or options for monitoring and managing the online video game play of the secondary users and associated notifications.

[0044] In some aspects, as the user of the electronic communication device **180** (e.g., the primary user) attempts to connect over the network **130** to the video gaming server **140**, the video gaming server **140** may request a verification of the identity (e.g., username/password) of the user (this username/password combination would have set up when setting up the account). The video gaming server **140** may verify the identity of the user of the electronic communication device **180** (e.g., by comparing the username/password data entered by the user of the electronic communication device **180** into the login interface against username/password data stored (e.g., in the memory of the video gaming server **140**) in association with the profile of the primary user). If the entered username/password match the stored username/password, the video gaming server **110** may then retrieve the information stored in the profile of the primary user and enable the user to log in to the video gaming server **140**.

[0045] Notably, in some aspects, the profile of the primary user may include electronic data representative of the access level of each user associated with the "family" account created by the primary user. For example, the access level may be such that the primary user and another person associated with the account and designated by the primary user have privileges to set up various permissions, restrictions, and monitoring/managing options with respect to the secondary users associated with the account. On the other hand, the access level may also restrict the ability of the secondary users associated with the account to view and/or modify the permissions, restrictions and/or monitoring/managing options set up by the primary user of the account.

[0046] In certain implementations, as will be discussed in more detail below, the video gaming monitoring application **190** is configured to generate a notification **128** for a primary user (e.g., parent, guardian, etc.) of the electronic communication device **180** of an online risk encountered by a secondary user (e.g., a child) playing the video gaming device **110** and linked by a common account (e.g., a "family" account) to the primary user of the electronic communication device **180**. In one aspect, as discussed below, the notification **128** displayed by the video gaming monitoring application **190** within the user interface **185** and on a display **182** of the electronic communication device **180** presents the primary user of the electronic communication device **180** with one or more options (displayed by the video gaming monitoring application **190**) for responding to the online risk encountered by the secondary user of the video gaming device **110** while playing a video game online with other players.

[0047] With reference to FIG. **2**, an exemplary video gaming server **140** configured for use with exemplary systems and devices described herein includes a control circuit **310** including a programmable processor (e.g., a microprocessor or a microcontroller) electrically coupled via a connection **315** to a memory **320** and via a connection **325** to a power supply **330**. The control circuit **310** can comprise a fixed-purpose hard-wired platform or can comprise a partially or wholly programmable platform, such as a microcontroller, an application specification integrated circuit, a field programmable gate array, and so on. These architectural options are well known and understood in the art and require no further description here.

[0048] The control circuit **310** can be configured (for example, by using corresponding programming stored in the memory **320** as will be well understood by those skilled in the art) to carry out one or more of the steps, actions, and/or functions described herein. In some embodiments, the memory **320** may be integral to the processor-based control circuit **310** or can be physically discrete (in whole or in part) from the control circuit **310** and is configured non-transitorily store the computer instructions that, when executed by the control circuit **310**, cause the control circuit **310** to behave as described herein. (As used herein, this reference to "non-transitorily" will be understood to refer to a non-ephemeral state for the stored contents (and hence excludes when the stored contents merely constitute signals or waves) rather than volatility of the storage media itself and hence includes both non-volatile memory (such as read-only memory (ROM)) as well as volatile memory (such as an erasable programmable read-only memory (EPROM))). Accordingly, the memory and/or the control unit may be referred to as a non-transitory medium or non-transitory computer readable medium.

[0049] In the embodiment illustrated in FIG. **2**, the control circuit **310** of the video gaming server **140** is also electrically coupled via a connection **335** to an input/output **340** that can receive signals from and/or send signals to, for example, the video gaming device **110**, risk module **150**, electronic communication device **180**, etc.

[0050] The processor-based control circuit **310** of the video gaming server **140** shown in FIG. **2** is electrically coupled via a connection **345** to a user interface **350**, which may include inputs **370** (e.g., keyboard, mouse, touch screen, etc.) that permit an operator of the video gaming server **140** to manually control the video gaming server **140** by inputting commands via touch button operation and/or voice commands and/or via a physically connected device. In some embodiments, the user interface **350** of the video gaming server **140** may also include a speaker **360** that provides audible feedback (e.g., notifications, alerts, etc.) to the operator of the video gaming server **140**. It will be appreciated that the performance of such functions by the processor-based control circuit **310** of the video gaming server **140** is not dependent on a human operator, and that the control circuit **310** of the video gaming server **140** may be programmed to perform such functions without a human operator.

[0051] As pointed out above, the risk module **150** and the video gaming server **140** may be implemented as separate, stand-alone devices as shown in FIG. **1**. FIG. **2** shows an alternative embodiment where the video gaming server **140** incorporates the risk module **150**. In the embodiment illustrated in FIG. **2**, the risk module **150** is coupled to the processor-based control circuit **310** of the video gaming server **140** via a connection **355**. Like the risk module **150** of FIG. **1**, the risk module **150** of FIG. **2** includes or is coupled to a generative AI component **160** and a child safety machine learning model **170**.

[0052] As pointed out above, the generative AI component **160** and the child safety machine learning model **170**, in combination, are trained to detect and address potential risks or threats to the safety of secondary users (e.g., children) associated with a given account set up by a primary user. For example, the child safety machine learning model **170** may be designed to analyze text-based and voice-based messages received or sent by a secondary user while using the video gaming device **110** to play online, and to detect content or activities that may be harmful or inappropriate for the secondary user (e.g., by being explicit, violent, threatening, etc.). In one aspect, to maximize the effectiveness of the risk module **150**, the generative AI component **160** and the child safety machine learning model **170** are specifically tailored for each of the individual "family" accounts set up by primary users of the video gaming monitoring application **190** by virtue of being trained only on data generated by the online interactions of the secondary users associated with each individual "family" account.

[0053] FIG. **3** illustrates an exemplary electronic communication device **180** (in this case, a mobile phone) with an exemplary user interface **185** and main menu **186** of the user interface **185** generated by a video gaming monitoring application **190** executable by the electronic communication device **180** being displayed on a screen **182** of the electronic communication device **180**. As mentioned above, the video gaming monitoring application **190** permits the primary user of the electronic communication device **180** to monitor and/or manage the activity and online interactions of a secondary user linked to an account of the primary user while the secondary user plays video games on the video gaming device **110**.

[0054] In the embodiment shown in FIG. **3**, the graphical user interface **185** includes multiple interactive graphical icons **122** and **124**, each of which displays a status **121**, an avatar **123**, and a name **125** of each secondary user (e.g., a child of the primary user) associated with a "family" account set up by the primary user. In this example, generic names Child 1 and Child 2 are used within the graphical icons **122** and **124**, and a generic name "Video Game **1**" for the video game title Child 1 is currently playing while Child 2 is Offline. It will be appreciated that the information displayed within the graphical icons **122**, **124** is shown by way of example only, and the icons **122**, **124** may include additional or alternative information in other embodiments.

[0055] The exemplary graphical user interface **185** depicted in FIG. **3** further includes informational fields **126**, **127** and **128**. In the illustrated example, informational field **126** shows the name of the secondary user (in this case, Child 1) and indicates the status of the secondary user (in this case, Child 1 is currently playing a video game). The interactive informational field **127** of the main menu **186** of the graphical user interface **185** shows the name of (and, optionally, a picture/photo/graphic associated with) the video game (in this case, "Video Game 1") being played by Child 1, and avatars of online players who are playing Video Game 1 together with Child 1. In this example, three avatars are visible within the informational field **127** and a "+5" icon indicates that five additional players are also playing. Informational field **128** shows a notification of online risks being encountered by Child 1 (in this case, the notification is "Your child may be interacting with a stranger"). It will be appreciated that the informational fields **126**, **127**, and **128** displayed within the graphical user interface **185** are shown by way of example only, and that the user interface **185** may include additional or alternative informational fields in other embodiments.

[0056] In the embodiment illustrated in FIG. **3**, the information fields **127** and **128** of the user interface **185** are interactive and permit the primary user to tap/click on them to bring up sub-menus associated with the information fields **127** and **128**. For example, if the primary user were to interact with the informational field **128** displaying the notification that Child 1 may have encountered a stranger (i.e., a player that is not in the friends list of Child 1), the graphical user interface **185** is configured to generate an exemplary interactive menu **131** visible in FIG. **4**. In the embodiment illustrated in FIG. **4**, the interactive menu **131** is displayed within the graphical user interface **185** as an overlay that overlays a portion of the main menu **186** of the graphical user interface **185** (shown in FIG. **3**) such that portions of the main menu **186** remain visible.

[0057] The exemplary menu **131** of the graphical user interface **185** shown in FIG. **4** includes an interactive informational field **132**, which, in this example, identifies the avatar **133** and screen name **134** (the generic screen name "Online Player 1" is used in this example) of the potential threat, and provides a text-based explanation **135** of the potential risk (in this case, pointing out that the user "Online Player 1" is a "Potential Bad Actor"). In the illustrated embodiment, to expand the informational field **132** and obtain more information regarding the gamer Online Player 1, the primary user of the gaming monitoring application **190** may interact with (e.g., tap, click, etc.) any portion of the information field **132**, or may interact with (e.g., tap, click, etc.) an interactive graphical icon **136** to expand the informational field **132**.

[0058] With reference to FIG. **4**, the exemplary informational field **132** further includes an informational field **137**, which may provide a text-based explanation of the detected potential online risk, i.e., "Your child is interacting with an account that users have tried to report multiple times," and indicate what the primary user can do to address this potential online risk, i.e., "You can control who your child can interact with in your parental control settings." In the illustrated embodiment, the interactive menu **131**, in addition to informing the primary user of the potential online risks to the secondary user, presents the primary user with one or more options for responding to the online risk by providing three separate interactive icons/buttons that the primary user may interact with.

[0059] For example, the exemplary menu **131** shown in FIG. **4** includes an interactive icon **138***a* that permits the primary user to block the potential bad actor Online Player 1. In other words, to block the gamer Online Player 1 from interacting (e.g., messaging, chatting, etc.) with Child 1 when Child 1 plays online, the primary user would interact with (e.g., press or click) icon/button **138***a* (which, in this example, is labeled "Block This Player"). In the illustrated embodiment, when the primary user interacts with the icon **138***a* (i.e., "Block This Player"), the graphical user interface **185** is configured to generate an exemplary interactive menu **141** visible in FIG. **5**. In the embodiment of FIG. **5**, the interactive menu **141** is displayed within the graphical user interface **185** as an overlay over a portion of the main menu **186** of the graphical user interface **185** (shown in FIG. **3**) such that portions of the main menu **186** remain visible.

[0060] In the embodiment illustrated in FIG. **4**, the exemplary menu **131** shown in FIG. **4** includes an interactive icon **138***b* (in this example, named "Leave the Chat"), which permits the primary user to leave/close the informational field **132** without taking any action with respect to the gamer Online Player 1. In some embodiments, the interactive icon **138** named "Leave the Chat" may permit the primary user to force Child 1 to leave the current interaction with the potential bad actor (i.e., disconnect Child 1 from the current interaction). The exemplary menu **131** of FIG. **4** further includes an interactive icon/button **138***c* (in this example, called "Edit Settings," which permits the primary user to edit the settings with respect to monitoring and/or management of online risk detection with respect to secondary user (i.e., Child 1).

[0061] Similar to the exemplary menu **131** of FIG. **4**, the exemplary menu **141** of the graphical user interface **185** shown in FIG. **5** includes an interactive informational field **142**, which, identifies the avatar **143** and screen name **144** of the online gamer that presents potential risk and provides a text-based explanation **145** of the potential threat (in this case, pointing out that the gamer "Online Player 1" is a "Potential Bad Actor"). The informational field **142** of menu **141** may be expanded to obtain more information regarding the gamer Online Player 1 by interacting with any portion of the information field **142**, or by interacting with an interactive graphical icon **146** to expand the informational field **142**.

[0062] The exemplary menu **141** shown in FIG. **4** further includes an interactive icon/button **148***a*, which permits the primary user to leave/close the menu **141** without blocking the gamer Online Player 1. In the illustrated embodiment, the interactive icon/button **148***a* displayed within the menu **141** is called "Cancel."

[0063] The exemplary menu **141** shown in FIG. **5** also includes an interactive icon/button **148***b*, which permits the primary user to block the gamer Online Player 1 from interacting with Child 1 when Child 1 plays online. In the illustrated embodiment, the interactive icon/button **148***b* displayed within the menu **141** is called "Block." The menu **141** shown in FIG. **5** further includes an informational field **147**, which, in this example, provides a text-based explanation of the effect of the "Block" action that may be taken by the primary user by interacting with the icon **148***b*. In this example, the informational field **147** indicates that "Blocking this player prevents them from interacting with Child 1 through Parties and Messages."

[0064] In one aspect, when the primary user interacts with the icon **148***b* (i.e., "Block"), the graphical user interface **185** is configured to generate an exemplary interactive menu **151** visible in FIG. **6**. In the embodiment illustrated in FIG. **6**, the interactive menu **151** is displayed within the graphical user interface **185** as an overlay that overlays a portion of the main menu **186** of the graphical user interface **185** (shown in FIG. **3**) such that portions of the main menu **186** remain visible.

[0065] The exemplary menu **151** of the graphical user interface **185** shown in FIG. **6** includes an informational field **157**, which, in this example, provides a text-based explanation of the action that has been taken by the video gaming monitoring application **190** in response to the primary user pressing icon **148***b* within the menu **141** of FIG. **5**. In this example, the informational field **157** indicates that the action taken in this case is "Player Blocked." In the illustrated embodiment, the informational field **157** may also indicate further actions that the primary user can take within the video gaming monitoring application **190**. In this example, the informational field **157** provides a text-based message indicating that "You can also adjust this setting to prevent strangers from interacting with Child 1 in the future."

[0066] To that end, the menu **151** includes an interactive informational field **152**, which includes a text field **153** indicating that the setting that can be adjusted by interaction with the informational field **152** (in this example, "Who can interact with your child?"), as well as a text field **154** indicating the current setting that is currently set for the child (in this example, "Anyone"). In the illustrated embodiment, to expand the informational field **152** and obtain more information regarding the settings that may be applied by the primary user with respect to who the secondary user (e.g., Child 1) can interact while playing video games online, the primary user may interact with (e.g., tap, click, etc.) any portion of the information field **152**, or may interact with (e.g., tap, click, etc.) an interactive graphical icon **156** to expand the informational field **152**.

[0067] The exemplary menu **151** of FIG. **6** further includes an interactive icon/button **158**, which permits the primary user to close the menu **151** and return to the main menu **186** of the user interface **185**, which is illustrated in FIG. **3**. In the illustrated embodiment interactive icon/button **158** displayed within the menu **151** is called "Done."

[0068] FIG. **7** illustrates the exemplary user interface **185** generated by a video gaming monitoring application **190** in response to the primary user of the electronic communication device **180** interacting with the interactive informational field **127** showing the name of (i.e., Video Game 1) and/or a graphic associated with the video game being played by the secondary user (i.e., Child 1). FIG. **7** shows that, the interactive informational field **127**, when interacted with (e.g., clicked, tapped, etc.) by the primary user, provides the primary user of the electronic communication device **180** with further information regarding the online interactions of Child 1 with other gamers while Child 1 plays Video Game 1.

[0069] In particular, the exemplary user interface **185** shown in FIG. **7** includes an informational field **165** (in this example, named "Messages"), which shows both the names of the online gamers who sent messages to Child 1 and when each of the messages was sent to Child 1. In the example shown in FIG. **7**, the informational field **165** entitled Messages includes: an informational field **169***a* indicating that Online Player 1 sent a message to Child 1 one minute ago; an informational field **169***b* indicating that a gamer in the Friends list of Child 1 sent a message to a group including Child 1 thirty minutes ago; and an informational field **169***c* indicating that another gamer (i.e., "My Other Friends") in the Friends list of Child 1 sent a message to Child 1 one minute ago. In some embodiments, each of the informational fields **169***a*-**169***c* is interactive and, when interacted with by the primary user, displays the content of the online message associated with the one of the informational fields **169***a*-**169***c* the primary user interacted with.

[0070] The exemplary user interface **185** shown in FIG. **7** further includes an informational field **155** (in this example, named "Party"), which shows the list of online gamers Child 1 is currently playing Video Game 1 with (and with whom Child 1 may be messaging and/or chatting while playing Video Game 1). In the example shown in FIG. **7**, the main menu **186** of the graphical user interface includes interactive informational fields **159***a* and **159***b*, indicating that Child 1 is playing Video Game 1 with gamers Online Player 1 (visibly identified in informational field **159***a* as a "Potential Bad Actor" with respect to Child 1) and Online Player 2 (identified in informational field **159***b* as a "Friend" with respect to Child 1), respectively.

[0071] In the example shown in FIG. **7**, if the primary user were to interact with the interactive informational field **159***a* displaying the notification that user Online Player 1 is a Potential Bad Actor, the graphical user interface **185** is configured to generate an exemplary interactive menu **171**, which is visible in FIG. **8** and akin to the menu **131** depicted in FIG. **4**. In the embodiment illustrated in FIG. **8**, the interactive menu **171** is displayed within the graphical user interface **185** as an overlay that overlays a portion of the main menu **186** of the graphical user interface **185** (shown in FIG. **3**) such that portions of the main menu **186** remain visible.

[0072] In the embodiment illustrated in FIG. **8**, the interactive menu **171** includes an informational field **172**, which, in this example, identifies the avatar **173** and screen name **174** (in this example, Online Player 1) of the online gamer presenting a potential risk to Child 1. In addition, the exemplary menu **171** displays a notification **175** of the online risk being encountered by Child 1 (in this case, the notification is "Your child may be interacting with a stranger"), as well as several interactive icons/buttons **178***a*-**178***c,* which permit the primary user to respond to the notification **175** of an online risk to Child 1 displayed within the menu **171**.

[0073] For example, interactive icon **178***a* permits the primary user to block the potential bad actor Online Player 1. In other words, to block Online Player 1 from interacting with Child 1 when the Child 1 plays online, the primary user would interact with (e.g., press or click) the icon/button **178***a* (which, in this example, is labeled "Block This Player").

[0074] Interactive icon **178***b* of the exemplary menu **171** (in this example, named "Leave the Chat") permits the primary user to leave/close the menu **171** without taking any action with respect to Online Player 1. The exemplary menu **171** of FIG. **8** further includes an interactive icon/button **178***c* (in this example, called "Edit Settings," which permits the primary user to edit the settings with respect to monitoring and/or managing online risks encountered by the secondary user (i.e., Child 1). In the illustrated embodiment, the interactive menu **171** further includes an informational field **179** (in this example, called "Recently Played") that indicates the video games recently played by Child 1 (in this example, Video Game 1, Video Game 2, and Video Game 3).

[0075] FIGS. **3-8** discussed above describe various features and functions of the video gaming monitoring application **190** in the context of monitoring/managing online risks that a secondary user (e.g., Child 1, Child 2, etc.) associated with an account of the primary user (e.g., parent, guardian, etc.) of the electronic communication device **180**. FIGS. **9-11**, on the other hand, relate to exemplary activity reports that may be generated by the video gaming monitoring application **190** with respect to any of the secondary users, and presented to the primary user within the graphical user interface **185** on the screen **182** of the electronic communication device **180**.

[0076] For example, FIG. **9** shows the graphical user interface **185** displaying an exemplary informational field **191** including a gaming activity report **192***a* (in bar graph form, but any other form other than bar graph may be used) for the secondary user (in this example, Child 1) within a time interval selected by the primary user. In particular, the informational field **191** includes an interactive visual indicator **192***b* (called "Day") that allows the primary user to select one day as the chosen time interval, and an interactive visual indicator **192***c* (called "Week") that allows the primary user to select one week as the chosen time interval. It will be appreciated that, in some embodiments, the graphical user interface **185** may also include an interactive visual indicator that would permit the primary user to select one month or one year as the chosen time interval. In the example shown in FIG. **9**, the primary user has selected one day as the desired time interval and the day chosen by the primary user in this example is "Today, July 10."

[0077] In the embodiment illustrated in FIG. **9**, the informational field **191** includes a visual indicator **192***d* of the total time (in this example, 1 hour, 45 minutes) Child 1 played video games on the day (i.e., July 10) selected by the primary user. The informational field **191** further includes a visual indicator **192***e* indicating whether Child 1 has exceeded the maximum video game playing time permitted (in the settings of the gaming monitoring application **190**) for Child 1 by the primary user. In this case, the visual indicator **192** indicates that Child 1 has exceeded the maximum permissible daily video game playtime, which is reflected by the exclamation sign within a triangle and by a text-based message that says "Playtime Passed."

[0078] In the illustrated embodiment, to provide the primary user with more control over the time Child 1 spends on video games, the informational field **191** includes an interactive icon/button **192***f* (in this example, called "Notify Playtime"), which, when interacted with by the primary user, causes the video gaming monitoring application **190** to generate a visible (e.g., on the screen **182** of the electronic communication device **180**) and/or an audible (e.g., via a speaker of the electronic communication device **180**) notification that the maximum video game playtime permitted for Child 1 has been reached by Child 1. This, of course, would advantageously enable the primary user to inform Child 1 that Child 1 has to stop playing video games before Child 1 overplays the permitted playtime by 1 hour and 45 minutes, as in the example shown in FIG. **9**.

[0079] The informational field **191** of the exemplary graphical user interface **185** of FIG. **9** further includes an interactive icon/button **192***g* (in this example, called "Edit Playtime"), which, when interacted with by the primary user, causes the video gaming monitoring application **190** to generate, within the graphical user interface **185**, a menu that enables the primary user to decrease or increase the total time per day or per week that Child 1 is permitted to play video games on the video gaming device **110**, or to set specific time intervals (e.g., from 6pm to 7pm) when the Child 1 is permitted to play video games on a given day.

[0080] In the embodiment illustrated in FIG. **9**, the informational field **191** of the graphical user interface **185** of FIG. **9** further includes an informational field **192***h* (in this example, called "Daily Breakdown"), which includes a listing indicating the title of each video game played by Child 1 on the day selected by the primary user, as well as the total time Child 1 played each video game on the day selected by the primary user. For example, in the example shown in FIG. **9**, the informational field **192***h* shows two listings **192***i* and **192***j,* with listing **192***i* indicating that Child 1 played one game, i.e., Video Game 4 for 1 hour in one gaming session, and listing **192***j* indicating that Child 1 played another game, i.e., Video game 5 for 1 hour in another gaming session.

[0081] In the illustrated embodiment, each listing **192***i* and **192***j* is an interactive field, which, when interacted with (e.g., tapped, clicked, etc.) by the primary user, enables the primary user to see more detailed information with respect to either of the two gaming sessions listed under the informational field **192***h*. For example, to expand the listing **192***i* and obtain more information regarding the latest gaming session engaged in by Child 1, the primary user may interact with (e.g., tap, click, etc.) any portion of interactive field **192***i* or **192***j*, or may interact with (e.g., tap, click, etc.) a graphical icon **192***k* or **192***m* associated with interactive fields **192***i* and **192***j*, respectively.

[0082] FIG. **10** is similar to FIG. **9**. However, while FIG. **9** shows the graphical user interface **185** after the primary user interacted with the interactive icon **192***b* (i.e., Day), FIG. **10** shows the graphical user interface **185** after the primary user interacted with the interactive icon **192***c* (i.e., Week), with the week being chosen by the primary user in this being "July 2-July 8." As such, the exemplary graphical gaming activity report **192***a* (in chart form) indicates how many hours per each day of the week of July 2-July 8 Child 1 spent playing video games. In addition, in the illustrated example, gaming activity report **192***a* may indicate the playtime that does not exceed the maximum permitted daily playtime in one color (e.g., green) and the playtime that exceeds the maximum permitted daily playtime in another color (e.g., red).

[0083] In the embodiment illustrated in FIG. **10**, the informational field **191** includes a visual indicator **192***d* of the total time (in this example, 9 hours 48 minutes) Child 1 played video games during the week selected by the primary user. The exemplary visual indicator **192***e* of informational field **191** of FIG. **10** indicates the daily average time (in this example, 1 hour 24 minutes) spent by Child 1 playing video games.

[0084] Similarly to FIG. **9**, the graphical user interface **185** displayed in FIG. **10** includes an interactive icon/button **192***g* (in this example, called "Edit Playtime"), which, when interacted with by the primary user, causes the video gaming monitoring application **190** to generate, within the graphical user interface **185**, a menu that enables the primary user to decrease or increase the total time per day or per week that Child 1 is permitted to play video games on the video gaming device **110**, or to set specific time intervals (e.g., from 6pm to 7pm) when Child 1 is permitted to play video games on a given day. In the embodiment illustrated in FIG. **10**, the informational field **191** of the graphical user interface **185** also includes an informational field **192***h* (i.e. "Weekly Breakdown), which was discussed in detail with reference to FIG. **9**.

[0085] FIG. **11** illustrates an exemplary graphical user interface **185** that includes an informational field **193** displaying various monitoring/management settings options that may be viewed and/or adjusted by the primary user. The exemplary informational field **193** associated with the settings-related options (and called "Settings" in this example) includes a listing of various settings/restrictions that the primary user can set with respect to Child 1. In the example shown in FIG. **11**, the informational field **193** called "Settings" has three underlying listings **194***a*-**194***c*.

[0086] In particular, listing **194***a* is an interactive field, which, when interacted with (e.g., clicked, tapped, etc.) by the primary user, permits the primary user to view various "Communication and UGC" options with respect to Child 1. Notably, the exemplary interactive field **194***a* indicates the status of the Communication and UGC setting (i.e., "Not Restricted"). In the illustrated embodiment, to cause the graphical user interface **185** to expand the listing **194***a* and view and/or adjust the Communication and UGC settings for Child 1, the primary user may interact with any portion of the interactive field **194***a*, or with the graphical icon **194***e* associated with interactive field **194***d*. This would allow the primary user to, for example, change the "Not Restricted" setting of Communication and UGC to a more restricted setting based on one or more parameters specified by the primary user.

[0087] Generally speaking, in the context of video game content, "Communication" may refer to the interaction and exchange of information between online players within a video game or through external channels. This can include in-game chat, voice chat, messaging systems, and social features that enable players to connect and collaborate. As used herein, "UGC" stands for User-Generated Content, which refers to content created by players themselves rather than the game developers. UGC may include custom levels, mods, character skins, game modes, and other player-created assets that enhance or modify the game experience.

[0088] The exemplary listing **194***b* is an interactive field, which, when interacted with (e.g., clicked, tapped, etc.) by the primary user, permits the primary user to view various "Age Filtering for Online Content" options with respect to Child 1. In the illustrated embodiment, to cause the graphical user interface **185** to expand the listing **194***b* and view the Age Filtering for Online Content settings for Child 1, the primary user may interact with any portion of the interactive field **194***b*, or with the graphical icon **194***e* associated with interactive field **194***b*. This would allow the primary user to, for example, change the settings for the type of content (e.g., "Restrict/Don't Restrict," or "G-rated," "PG-rated," "R-rated," etc.) that Child 1 may access and/or be exposed to when playing online.

[0089] Listing **194***c* is an interactive field, which, when interacted with (e.g., clicked, tapped, etc.) by the primary user, permits the primary user to view various "Monthly Spending Limit" options with respect to Child 1. In the example illustrated in FIG. **11**, the primary user preset a monthly limit of $15 as the maximum amount Child 1 can spend per month on video games, in-game purchases, etc. In the illustrated embodiment, to cause the graphical user interface **185** to expand the listing **194***c* and view and/or modify (e.g., increase or decrease) the Monthly Spending Limit for Child 1, the primary user may interact with any portion of the interactive field **194***c*, or with the graphical icon **194***f* associated with interactive field **194***c*.

[0090] The exemplary graphical user interface **185** shown in FIG. **11** further includes various profile-related options that may be viewed and/or adjusted by the primary user. In FIG. **11**, the exemplary informational field **195** associated with various profile-related options (and called "Profile" in this example) includes a listing of various options/restrictions that the primary user can set with respect to Child 1. In the example shown in FIG. **11**, the informational field **195** called "Profile" has three underlying listings **196***a*-**196***c*.

[0091] In particular, listing **196***a* is an interactive field (in this example, called "Games"), which, when interacted with (e.g., clicked, tapped, etc.) by the primary user, permits the primary user to view the video games played by Child 1 within a primary user-selected time interval and to view the total time spent by Child 1 playing each video game. For example, to cause the graphical user interface **185** to expand the listing **196***a* and view each video game played by Child 1 on a given day and the total time spent by Child 1 playing each video game, the primary user may interact with any portion of interactive field **196***a*, or with a graphical icon **196***d* associated with interactive field **196***a*.

[0092] Listing **196***b* is an interactive field, which, when interacted with (e.g., clicked, tapped, etc.) by the primary user, permits the primary user to view the "Friends" list of Child 1. For example, if the primary user were to expand the listing **196***b* by tapping on the interactive field **196***b* or on the graphical icon **196***e* associated with the interactive field **196***b*, the graphical user interface **185** would generate a list of online gamers that Child 1 added as a friend.

[0093] Listing **196***c* is an interactive field, which, when interacted with (e.g., clicked, tapped, etc.) by the primary user, permits the primary user to view, set, and/or adjust various profile-related options specific to the online video game play by Child 1. For example, to cause the graphical user interface **185** to expand the listing **196***c* and view/set/adjust various options associated with the profile of Child 1, the primary user may interact with any portion of interactive field **196***c*, or with a graphical icon **196***f* associated with interactive field **196***c*.

[0094] FIG. **12** illustrates the exemplary user interface **185** generated by a video gaming monitoring application **190** in response to the primary user of the electronic communication device **180** interacting with the interactive informational field **196***c* called "Profile View." In the example shown in FIG. **12**, the exemplary graphical user interface **185** displays an informational field **197** associated with various profile-related options (and called "Privacy Settings" in this example) that the primary user can set with respect to Child 1. In some embodiments, the Profile View enables the primary user to view what information in the profile of Child 1 friends of Child 1 will see when viewing the profile of Child 1 online.

[0095] In the example shown in FIG. **12**, the informational field **197** has five underlying listings **198***a*-**198***e*. In particular, the listing **198***a* in FIG. **12** is an interactive field, which, when interacted with (e.g., clicked, tapped, etc.) by the primary user, permits the primary user to view options relating to "Who can see your child's friends" with respect to Child 1. In some aspects, such options may include "Anyone," "Friends Only," "Friends of Friends," "No One," etc. Notably, the exemplary interactive field **198***a* includes a text-based indicator **198***f* indicating the current setting of the "Who can see your child's friends" setting. In the example illustrated in FIG. **12**, the text-based indicator **198***f* displays "Friends Only," which means that only video game players included in the Friends list of Child 1 are able to see the Friends list of Child 1. In the illustrated embodiment, to cause the graphical user interface **185** to expand the listing **198***a* and view the list of names/avatars of the players that are on the Friends list of Child 1 and are thus permitted to view the Friends list of Child 1, the primary user may interact with any portion of the interactive field **198***a*, or with the graphical icon **198***g* associated with the interactive field **198***a*.

[0096] The listing **198***b* in FIG. **12** is an interactive field, which, when interacted with (e.g., clicked, tapped, etc.) by the primary user, permits the primary user to view options relating to "Who can see your child's games" with respect to Child 1. In some aspects, such options may include "Anyone," "Friends Only," "Friends of Friends," "No One," etc. Notably, the exemplary interactive field **198***b* includes a text-based indicator **198***h* indicating the current setting of the "Who can see your child's games" setting. In the example illustrated in FIG. **12**, the text-based indicator **198***h* displays "Friends Only," which means that only gamers included in the Friends list of Child 1 are able to see the list of video games played by Child 1. In the illustrated embodiment, to cause the graphical user interface **185** to expand the listing **198***b* and view the list of video games played by Child 1, the primary user may interact with any portion of the interactive field **198***b*, or with the graphical icon **198***i* associated with the interactive field **198***b*.

[0097] The listing **198***c* in FIG. **12** is an interactive field, which, when interacted with (e.g., clicked, tapped, etc.) by the primary user, permits the primary user to view options relating to "Who can see your child's online status" with respect to Child 1. In some aspects, such options may include "Anyone," "Friends Only," "Friends of Friends," "No One," etc. Notably, the exemplary interactive field **198***c* includes a text-based indicator **198***j* indicating the current setting of the "Who can see your child's online status" setting. In the example illustrated in FIG. **12**, the text-based indicator **198***j* displays "Friends Only," which means that only video game players included in the Friends list of Child 1 are able to see the online status of Child 1 (i.e., whether Child 1 is online or offline). In the illustrated embodiment, to cause the graphical user interface **185** to expand the listing **198***c* and view the online status of Child 1, the primary user may interact with any portion of the interactive field **198***c*, or with the graphical icon **198***k* associated with the interactive field **198***c*.

[0098] The listing **198***d* in FIG. **12** is an interactive field, which, when interacted with (e.g., clicked, tapped, etc.) by the primary user, permits the primary user to view options relating to "Who can ask to be your child's friend" with respect to Child 1. In some aspects, such options may include "Anyone," "Friends Only," "Friends of Friends," "No One," etc. Notably, the exemplary interactive field **198***d* includes a text-based indicator **198***m* indicating the current setting of the "Who can see ask to be your child's friend" setting. In the example illustrated in FIG. **12**, the text-based indicator **198***m* displays "Friends Only," which means that only video game players included in the Friends list of Child 1 are able to send a "Friend Request" to Child 1. In the illustrated embodiment, to cause the graphical user interface **185** to expand the listing **198***d* and view the list of names/avatars of the players that are friends of Child 1 and are thus permitted to send friend requests to Child 1, the primary user may interact with any portion of the interactive field **198***d*, or with the graphical icon **198***n* associated with the interactive field **198***d*.

[0099] The listing **198***e* in FIG. **12** is an interactive field, which, when interacted with (e.g., clicked, tapped, etc.) by the primary user, permits the primary user to view options relating to "Who can interact with your child" with respect to Child 1. In some aspects, such options may include "Anyone," "Friends Only," "Friends of Friends," "No One," etc. Notably, the exemplary interactive field **198***e* includes a text-based indicator **198***o* indicating the current setting of the "Who can interact with your child" setting. In the example illustrated in FIG. **12**, the text-based indicator **198***o* displays "Friends of Friends," which means that only video game players who are Friends of the players in the Friends list of Child 1 are able to interact with Child 1 online. In the illustrated embodiment, to cause the graphical user interface **185** to expand the listing **198***e* and view the list of names/avatars of the players that are Friends of Friends Child 1 and are thus permitted to interact with Child 1 online, the primary user may interact with any portion of the interactive field **198***e*, or with the graphical icon **198***p* associated with the interactive field **198***e*.

[0100] FIG. **13** illustrates the exemplary user interface **185** generated by a video gaming monitoring application **190** in response to the primary user of the electronic communication device **180** interacting with the interactive informational field **196***e* called "Who can interact with your child." In the example shown in FIG. **13**, the exemplary graphical user interface **185** displays an exemplary sub-menu **181** including a text-based indicator **183***a* depicting the name of this menu **181**, in this example, "Who can interact with your child." In addition, the exemplary sub-menu **181** includes a text-based indicator **183***b* indicating what kind of options are available to the primary user within the sub-menu **181**. In the example illustrated in FIG. **13**, the text-based indicator **183***b* informs the primary user that the menu **141** allows the primary user to "Choose who can invite your child to parties, initiate message, and send them game invitations."

[0101] The exemplary menu **181** of FIG. **13** of the user interface **185** of the video gaming monitoring application **190** includes an exemplary drop-down menu **187**, which, when interacted with (e.g., clicked, tapped, etc.) by the primary user, permits the primary user to view selectable options relating to which online video gamers can interact with Child 1 while Child 1 is playing a video game online. The drop-down menu **187** visible within the exemplary menu **181** of FIG. **13** shows the "Anyone" option as being selected, which means that any video game player who is online may interact with Child 1. Notably, in the embodiment illustrated in FIG. **13**, the graphical icon **183***c* associated with the exemplary drop-down menu **187** is oriented with its peak pointed downwardly to indicate that the drop-down menu **187** is not in an expanded state.

[0102] FIG. **14** illustrates the exemplary user interface **185** generated by the video gaming monitoring application **190** in response to the primary user of the electronic communication device **180** interacting with the drop-down menu **187** to expand the drop-down menu **187** and display all exemplary options available for selection by the primary user within the drop-down menu **187**. In the example shown in FIG. **14**, the options available for selection to the primary user include "Anyone" (which is currently actively selected), "Friends Only" (the selection of which by the primary user means that only video game players in the Friends list of Child 1 may interact with Child 1), and "No One" (the selection of which by the primary user means that none of the video game players who are online may interact with Child 1). Notably, in the embodiment illustrated in FIG. **14**, the graphical icon **183***c* associated with the exemplary drop-down menu **187** is oriented with its peak pointed upwardly to indicate that the drop-down menu **187** is in an expanded state.

[0103] FIG. **15** illustrates the exemplary user interface **185** generated by the video gaming monitoring application **190** in response to the primary user of the electronic communication device **180** interacting with the "Friends Only" option of the drop-down menu **187** to restrict the amount of online players who can interact with Child 1 and to change the "Anyone" option (which allows any video game player who is online to interact with Child 1) to the "Friends Only" option (which allows only video game players in the Friends list of Child 1 to interact with Child 1).

[0104] In the embodiment illustrated in FIG. **15**, the graphical user interface further includes interactive icons/buttons **189***a* and **189***b*. Icon/button **189***a* (called "Cancel" in this example) permits the primary user to cancel the substitution of the "Anyone" option with the "Friends Only" option and to return the drop-down menu **187** with the "Anyone" option still being selected as shown in FIG. **14**. On the other hand, icon/button **189***b* (called "Save" in this example) permits the primary user to enact/implement the substitution of the "Anyone" option with the "Friends Only" option and to return the drop-down menu **187** with the "Friends Only" option being selected. With the "Friends Only" option being saved by virtue of the user having interacted with the button **189***b*, the gaming monitoring application **190** will no longer permit any video game player who is online to interact with Child 1, but will instead only permit the players on the Friends list of Child 1 to interact with Child 1 while Child 1 is online and playing a video game.

[0105] FIG. **16** shows an exemplary notification **128** of an online risk presented to Child 1 while Child 1 is playing video games online, but this notification **128** is not displayed within the graphical user interface **185** discussed above, and is instead displayed to the primary user within the main menu **186** of the graphical user interface **185** on the screen **182** of the electronic communication device **180**. The exemplary notification **128** includes an informational field **129***a* that indicates the name of the secondary user that has encountered an online risk while playing a video game online (in this case, Child 1). In addition, the exemplary notification **128** includes an informational field **129***b* that indicates the type of online risk that Child 1 has encountered while playing a video game online (in this case, that Child 1 is "interacting with a potential bad actor). The exemplary notification **128** also includes an informational field **128***c* that indicates when Child 1 encountered a risk while playing a video game online (in this example, 1 minute ago).

[0106] FIG. **17** shows an exemplary embodiment of an exemplary method **200** of monitoring and/or managing online video game play of one or more users. The method **200** includes monitoring online activities of a first user (e.g., a secondary user such as a child) that is operating a video gaming device **110** to play a video game online (step **210**).

[0107] The method **200** further includes determining that the first user has encountered an online risk while operating the video gaming device (step **220**). As pointed out above, this determination can be made by a risk module **150**, which may be coupled to or incorporated into the video gaming device **110** or the video gaming server **140** as discussed above. As also discussed above, the risk module **150** may include or be coupled to a generative AI component **160** and a child safety machine learning model **170**, which are trained to detect and address potential risks or threats to the safety of secondary users (e.g., children) associated with a given account set up by a primary user. For example, the child safety machine learning model **170** may be designed to analyze text-based and voice-based messages received or sent by a secondary user while using the video gaming device **110** to play online, and to detect content or activities that may be harmful or inappropriate for the secondary user (e.g., by being explicit, violent, threatening, etc.).

[0108] In the illustrated embodiment, the method **200** includes sending a notification of the online risk detected by the risk module **150** to have been encountered by the first user (e.g., a secondary user such as Child 1) to a second device (e.g., electronic communication device **180**) of a second user (e.g., a primary user of the online gaming monitoring application **190** such as a parent, guardian, etc.) (step **230**). As pointed out above, the first user and second users are associated with each other by a common account, and the notification **128** may be visually displayed to the second user on a display screen **182** of the electronic communication device **180** of the second user, and may be interactive to enable the second user to select one or more options for responding to the online risk encountered by the first user.

[0109] In view of the foregoing, the systems, graphical user interfaces, applications, and methods described in this application enable parents, guardians, and the like not only to easily and effectively monitor and manage the video game activity of their children, but to also be aware of, and alleviate any active risks that their children may be exposed online while playing video games with other gamers. As such, the safety of children while playing online is increased.

[0110] In some embodiments, one or more of the embodiments, methods, approaches, schemes, and/or techniques described above may be implemented in one or more computer programs or software applications executable by a processor-based apparatus or system. By way of example, such processor-based system may comprise a smartphone, tablet computer, virtual reality (VR), augmented reality (AR), or mixed reality (MR) system, entertainment system, game console, mobile device, computer, workstation, gaming computer, desktop computer, notebook computer, server, graphics workstation, client, portable device, pad-like device, communications device or equipment, etc. Such computer program(s) or software may be used for executing various steps and/or features of the above-described methods, schemes, and/or techniques. That is, the computer program(s) or software may be adapted or configured to cause or configure a processor-based apparatus or system to execute and achieve the functions described herein. For example, such computer program(s) or software may be used for implementing any embodiment of the above-described methods, steps, techniques, schemes, or features. As another example, such computer program(s) or software may be used for implementing any type of tool or similar utility that uses any one or more of the above-described embodiments, methods, approaches, schemes, and/or techniques. In some embodiments, one or more such computer programs or software may comprise a VR, AR, or MR application, communications application, object positional tracking application, a tool, utility, application, computer simulation, computer game, video game, role-playing game (RPG), other computer simulation, or system software such as an operating system, BIOS, macro, or other utility. In some embodiments, program code macros, modules, loops, subroutines, calls, etc., within or without the computer program(s) may be used for executing various steps and/or features of the above-described methods, schemes and/or techniques. In some embodiments, such computer program (s) or software may be stored or embodied in a non-transitory computer readable storage or recording medium or media, such as a tangible computer readable storage or recording medium or media. In some embodiments, such computer program(s) or software may be stored or embodied in transitory computer readable storage or recording medium or media, such as in one or more transitory forms of signal transmission (for example, a propagating electrical or electromagnetic signal).

[0111] Therefore, in some embodiments the present invention provides a computer program product comprising a medium for embodying a computer program for input to a computer and a computer program embodied in the medium for causing the computer to perform or execute steps comprising any one or more of the steps involved in any one or more of the embodiments, methods, approaches, schemes, and/or techniques described herein. For example, in some embodiments the present invention provides one or more non-transitory computer readable storage mediums storing one or more computer programs adapted or configured to cause a processor-based apparatus or system to execute steps comprising any one or more of the embodiments, methods, approaches, schemes, and/or techniques described herein.

[0112] While the invention herein disclosed has been described by means of specific embodiments and applications thereof, numerous modifications and variations could be made thereto by those skilled in the art without departing from the scope of the invention set forth in the claims.

1. A method comprising:

monitoring online activities of a first user that is operating a first device, wherein the first device is a video gaming device;

determining that the first user has encountered an online risk while operating the video gaming device;

sending a notification of the online risk to a second device operated by a second user; and

receiving, in response to sending the notification, a command from the second device operated by the second user to perform one or more of:

blocking a third user from interacting with the first user,

reporting the third user for inappropriate behavior,

flagging the third user for further review, and

sending a message from the second user to the third user;

2-20. (canceled)

21. The method of claim **1**, wherein the online risk comprises an interaction of the first user with the third user, the interaction including at least one of a message, an image, a video, or an audio.

22. The method of claim **1**, wherein the determining that the first user has encountered an online risk comprises:

detecting that an interaction between the third user and the first user violates one or more permissions or restrictions.

23. The method of claim **1**, wherein the determining that the first user has encountered an online risk comprises:

feeding information relating to online activity of the first user while the first user is operating the video gaming device into an artificial intelligence model trained to recognize the online risk to the first user.

24. The method of claim **1**, wherein the notification further includes a description of the online risk or content associated with the online risk.

25. The method of claim **1**, wherein the notification is transmitted from a video gaming server to a mobile application installed on the second device, and wherein the notification is displayed within a graphical interface generated by the mobile application.

26. The method of claim **1**, wherein the command is other than a command to retrain a model, block the first user, or ignore the online risk.

27. A system comprising:

one or more computer processors; and

one or more non-transitory computer-readable media that store instructions which, when executed, cause the one or more computer processors to perform operations comprising:

monitoring online activities of a first user that is operating a first device, wherein the first device is a video gaming device;

determining that the first user has encountered an online risk while operating the video gaming device;

sending a notification of the online risk to a second device operated by a second user; and

receiving, in response to sending the notification, a command from the second device operated by the second user to perform one or more of:

blocking a third user from interacting with the first user,

reporting the third user for inappropriate behavior,

flagging the third user for further review, and

sending a message from the second user to the third user.

28. The system of claim **27**, wherein the online risk comprises an interaction of the first user with the third user, the interaction including at least one of a message, an image, a video, or an audio.

29. The system of claim **27**, wherein the determining that the first user has encountered an online risk comprises:

detecting that an interaction between the third user and the first user violates one or more permissions or restrictions.

30. The system of claim **27**, wherein the determining that the first user has encountered an online risk comprises:

feeding information relating to online activity of the first user while the first user is operating the video gaming device into an artificial intelligence model trained to recognize the online risk to the first user.

31. The system of claim **27**, wherein the notification further includes a description of the online risk or content associated with the online risk.

32. The system of claim **27**, wherein the notification is transmitted from a video gaming server to a mobile application installed on the second device, and wherein the notification is displayed within a graphical interface generated by the mobile application.

33. The system of claim **27**, wherein the command is other than a command to retrain a model, block the first user, or ignore the online risk.

34. A non-transitory computer readable storage medium storing one or more computer instructions which, when executed by one or more computer processors, cause the one or more computer processors to perform operations comprising:

monitoring online activities of a first user that is operating a first device, wherein the first device is a video gaming device;

determining that the first user has encountered an online risk while operating the video gaming device;

sending a notification of the online risk to a second device operated by a second user; and

receiving, in response to sending the notification, a command from the second device operated by the second user to perform one or more of:

blocking a third user from interacting with the first user,

reporting the third user for inappropriate behavior,

flagging the third user for further review, and

sending a message from the second user to the third user.

35. The medium of claim **35**, wherein the determining that the first user has encountered an online risk comprises:

detecting that an interaction between the third user and the first user violates one or more permissions or restrictions.

36. The medium of claim **35**, wherein the determining that the first user has encountered an online risk comprises:

feeding information relating to online activity of the first user while the first user is operating the video gaming device into an artificial intelligence model trained to recognize the online risk to the first user.

37. The medium of claim **35**, wherein the notification further includes a description of the online risk or content associated with the online risk.

38. The medium of claim **35**, wherein the notification is transmitted from a video gaming server to a mobile application installed on the second device, and wherein the notification is displayed within a graphical interface generated by the mobile application.

39. The medium of claim **35**, wherein the command is other than a command to retrain a model, block the first user, or ignore the online risk.

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